

“E-COMMERCE SALES ANALYTICS DASHBOARD”

A

PROJECT REPORT

*Submitted in partial fulfillment of the requirement for the
award of Degree of*

Bachelor of Computer Applications

Submitted to

310-Cimage Professional College



Submitted By: KAHKASHA NOOR

Registration No: 23303310148

BCA – 6th Sem

Session-2023-2026

Under Joint Guidance of

**Mr. Niraj Kumar Singh &
Er. Pankaj Chandra Sinha**

[Asst. Professor]

Computer Science &
Information Technology

DATE:

CERTIFICATE

This is to certify that the work embodies in this project entitled, “**E-Commerce Sales Analytics Dashboard**”, being submitted by **KAHKASHA NOOR** (Registration Number: 23303310148) in partial fulfillment of the requirement for the award of “**Bachelor of Computer Applications**” to **ARYABHATTA KNOWLEDGE UNIVERSITY, PATNA** during the academic year 2023-26 is a record of bonafide piece of work, carried out by him/her under our/my supervision and guidance in the “**Department of Computer Science & Information Technology**”, Cimage Professional College, Patna.



Dr. Neeraj Agrawal
Director

Dr. Amit Kumar Shukla
Head of Department -IT

Date: _____

APPROVAL CERTIFICATE

The project report entitled “**E-Commerce Sales Analytics Dashboard**”, being submitted by **Kahkasha Noor** (Registration Number: **23303310148**) has been examined by us and is hereby approved for the award of degree “**Bachelor of Computer Applications**”, for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, the opinion expressed or conclusion drawn therein, but approve the project only for the purpose for which it has been submitted.



(Internal Examiner)

(External Examiner)

Date:

Date:



Date: _____

DECLARATION

I, **Kahkasha Noor** hereby declare that the work, which is being presented in the project report entitled “**E-Commerce Sales Analytics Dashboard**”, partial fulfillment of the requirements for the award of degree of “**Bachelor of Computer Applications**”, submitted in the department of **Computer Science & Information Technology (Cimage Professional College)** is an authentic record of my own work carried under the joint guidance of **Niraj Kumar Singh Sir & Pankaj Chandra Sinha Sir**. I have not submitted the matter embodied in this report for the award of any other degree.

Kahkasha Noor
Registration No : 23303310148
BCA 6th Sem
Session-2023-2026

ACKNOWLEDGEMENT

“A journey is easier when you travel together. Interdependence is certainly more valuable than independence.”

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“The completion of any project depends upon the cooperation, coordination, and combined efforts of several resources of knowledge, inspiration, and energy”.

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Name: Kahkasha Noor
Registration No : **23303310148**
BCA 6th sem
Session-2023-2026

ABSTRACT

The E-Commerce Sales Analytics Dashboard is a web-based analytics tool designed to give insights into the sales data of e-commerce companies. Companies nowadays produce huge amounts of transaction data every single day.

It is hard and tedious to analyze all the data by oneself; therefore, the quality of decisions may suffer from this process.

The purpose of this project is to turn sales data into valuable business insights with the help of visualization, reporting, and dashboarding tools.

This project uses such technologies as Python, Streamlit, SQLite, Pandas, Plotly, and ReportLab. The project uses role-based authentication, activity logging, generating PDF reports, user management features, and data visualization tools.

With the help of the dashboard, business users and administrators can make better and more informed decisions based on the analytical information provided.

Keywords:

E-Commerce, Analytics Dashboard, Data Visualization, Business Intelligence, Streamlit, Python, SQLite, Sales Analytics

CONTENTS

REPORT	i
INTRODUCTION	1
OBJECTIVE	1
PROJECT CATEGORY DATA ANALYTICS	10
ADVANTAGES OF DATA ANALYTICS	10
TOOLS/PLATFORM REQUIRED	11
HARDWARE & SOFTWARE REQUIREMENTS	12
PROBLEM DEFINITION	13
REQUIREMENT SPECIFICATIONS	13
PROJECT PLANNING & SCHEDULING	14
GANTT CHART	16
PERT CHART	17
SYSTEM ANALYSIS	18
IDENTIFICATION OF NEED	18
FEASIBILITY STUDY	19
TECHNICAL FEASIBILITY	19
ECONOMICAL FEASIBILITY	20
OPERATIONAL FEASIBILITY.....	21
ANALYSIS.....	22
DFD	23
MODULE DESCRIPTION	24
DATA STRUCTURE SNAPSHOT	25
LIST OF REPORTS	26
SCOPE AND FUTURE ENHANCEMENT	27
SYSTEM ANALYSIS	27

SYSTEM DESIGN.....35

IMPLEMENTATION METHODOLOGY.....42

TESTING 53

RESULTS AND DISCUSSION 65

SECURITY MECHANISM83

SCOPE AND FUTURE ENHANCEMENTS92

CONCLUSION 101

BIBLIOGRAPHY 105

SOURCE CODE107

APPENDIX 121

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The rapid growth of digital technology has transformed the way businesses operate and manage information. E-commerce platforms generate a large amount of sales data every day through customer purchases, transactions, product browsing, and payment activities. This data contains valuable information that can help organizations understand customer behavior, monitor business performance, and improve decision-making processes.

Traditional methods of analyzing sales data using spreadsheets and manual calculations are time-consuming and prone to errors. Businesses require intelligent systems capable of processing large volumes of data and presenting meaningful insights in an easy-to-understand format.

The E-Commerce Sales Analytics Dashboard is a data analytics solution developed to analyze e-commerce sales data efficiently. The system provides interactive dashboards, graphical reports, user authentication, activity logging, and PDF report generation. The dashboard enables business managers and decision-makers to monitor sales performance, profit trends, regional performance, and customer insights through visual representations.

The project is developed using Python, Streamlit, Pandas, Plotly, SQLite, and ReportLab. These technologies provide a robust platform for building a user-friendly and efficient analytics system.

The primary purpose of this project is to transform raw sales data into meaningful business intelligence that supports strategic decision-making and improves organizational performance.

1.2 OBJECTIVES

The main objectives of the E-Commerce Sales Analytics Dashboard include:

- To conduct efficient sales data analysis.
- To create a visual representation of sales.
- To track the progress in terms of profit and business growth.
- To have a secure user authentication mechanism.

- To have role-based access for users.
- To offer PDF report creation functionality.
- To log user activities.
- To help decision-making process using analysis.
- To offer an interactive and user-friendly dashboard.

1.3 PROJECT CATEGORY – DATA ANALYTICS

E-Commerce Sales Analytics Dashboard falls under the category of Data Analytics and Business Intelligence.

Data Analytics refers to the practice of gathering, processing, analysis, and visualization of data in order to draw out meaningful information and make informed decisions. Businesses implement analytics systems to discover trends, patterns, possibilities, and problems in their businesses.

The project integrates ideas from the following domains:

- Data Analytics
- Business Intelligence
- Data Visualization
- Database Management
- Web Application Development
- Report Generation
- User Management Systems

The dashboard functions as a business intelligence system by generating meaningful reports from the raw data of sales.

1.4 BLESSINGS OF DATA ANALYTICS

There are many blessings of data analytics for an organization.

- **Effective Decision Making**
Organizations can make effective decisions using data analytics, not their assumption.

- **Understanding of Customer Preferences**
The analytics system assists in knowing the customer's preferences and behavior.

- Greater Efficiency

The automated data analytics process saves time and energy.

- Better Business Performance

The company can analyze its performance by using KPIs.

- Avoiding Risks

Data analytics will help you to recognize possible business risk and challenges.

- Effective Resource Distribution

Organizations can effectively distribute the resources using performance metrics.

- Dashboards System

This system helps in getting the important information about your business instantly.

1.5 TOOLS / PLATFORM REQUIRED

The project utilizes the following tools and platforms:

- Programming Language:

Python

Framework:

Streamlit

- Libraries:

Pandas

Plotly

Matplotlib

NumPy

ReportLab

- Database:

SQLite

- Version Control:
Git and GitHub
- Deployment Platform:
Streamlit Cloud
- Development Environment:
Visual Studio Code
- Operating System:
Windows 10 / Windows 11

1.6 HARDWARE & SOFTWARE REQUIREMENTS

Hardware Requirements

- Processor:
Intel Core i3 or higher
RAM:
4 GB minimum
8 GB recommended
- Storage:
10 GB free disk space
Display:
1366 × 768 resolution or higher
- Internet Connection:
Required for deployment and updates
Software Requirements
- Operating System:
Windows 10 / Windows 11
- Programming Language:
Python 3.12+
- Framework:
Streamlit

- Database:
SQLite
- Code Editor:
Visual Studio Code
- Browser:
Google Chrome / Mozilla Firefox
- Version Control:
Git

1.7 PROBLEM STATEMENT

Modern companies engaged in e-commerce create huge amounts of sales data via transactions carried out on the Internet. Manual handling and analyzing of such data become increasingly hard as the amount of data increases.

The problems arising before organizations are the following:

- The challenge of tracking sales performance.
- The time-consuming process of manual analysis.
- The lack of proper data visualization.
- Late report creation.
- Possible mistakes in calculations.
- Bad decision-making.

A unified analytical tool that can handle sales data and provide important information via interactive reports is required.

The E-Commerce Sales Analytics Dashboard is meant to resolve the above issues.

1.8 REQUIREMENTS SPECIFICATIONS

Functional Requirements

- User Login
- User Logout
- Access Control Based on Roles
- Dashboard Displaying Data

- Sales Analytics
- Profit Analytics
- User Management
- Logging Activities
- Generating PDF Reports

Non-functional Requirements

- Security
- Reliability
- Performance
- Scalability
- Maintainability
- Usability

Data Requirements

- Sales Data
- User Data
- Activity Logs
- Generated Reports

1.9 PROJECT PLANNING & SCHEDULING

Project planning is an important task which involves the systematic completion of project activities within the designated period of time.

The process of development included several stages.

Stage 1 – Requirement Analysis

Activities:

- Requirement Gathering
- Problem Definition
- Goal Setting

Time Frame:

2 Weeks

Stage 2 – System Designing

Activities:

- Architecture designing
- Database designing
- Interface designing

Time Frame:

2 Weeks

Stage 3 – Development

Activities:

- Development of login module
- Development of dashboard
- Database integration
- Implementation of user management

Time Frame:

4 Weeks

Stage 4 – Testing

Activities:

- Unit testing
- Integration testing
- Error detection and correction

Time Frame:

2 Weeks

Stage 5 – Deployment

Activities:

- GitHub upload
- Deployment on Streamlit
- Final verification

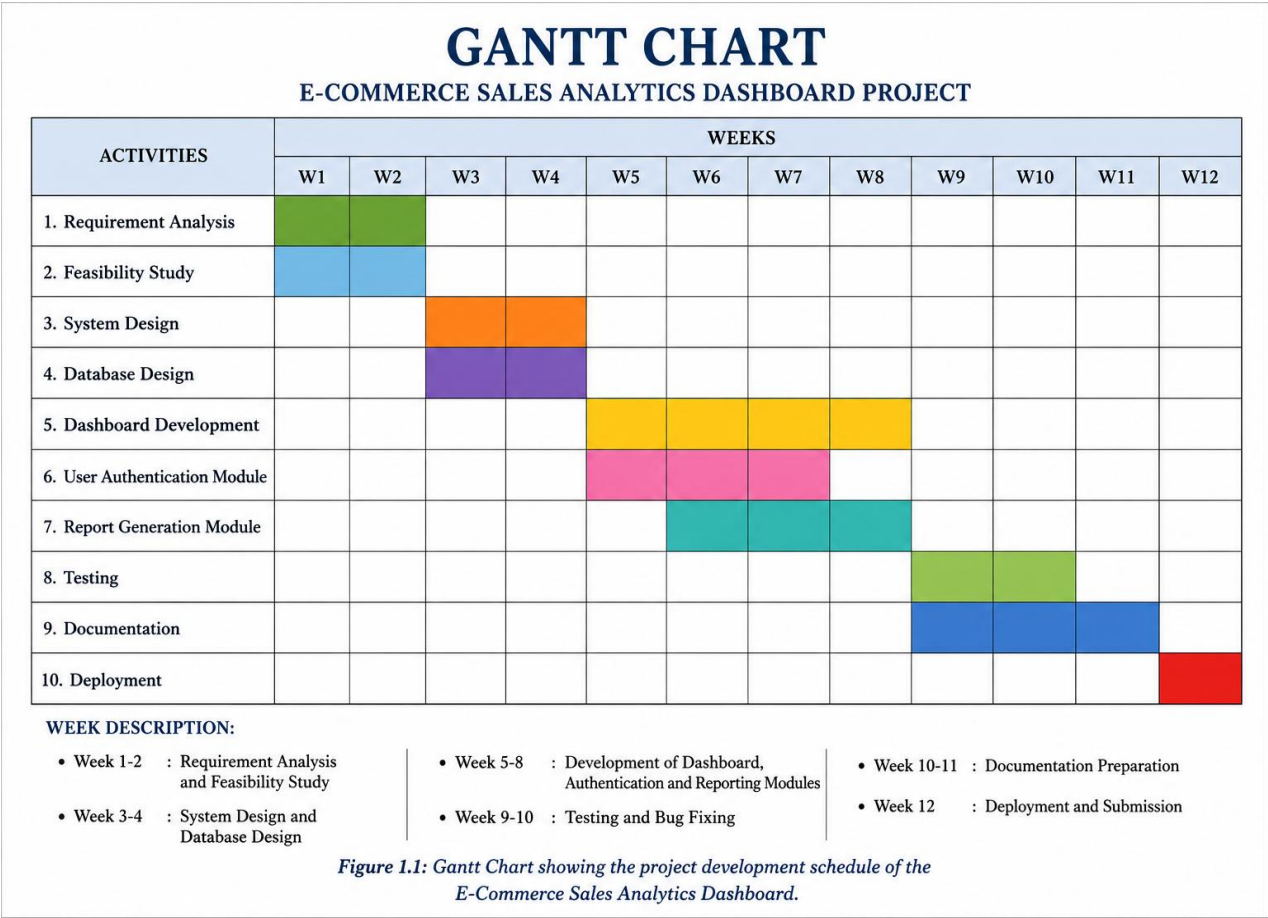
Time Frame:

1 Week

1.10 GANTT CHART

Project Activity Week

- Requirement Analysis 1–2
- System Design 3–4
- Database Design 3–4
- Development 5–8
- Dashboard Creation 5–8
- Testing 9–10
- Documentation 10–11
- Deployment 12



1.11 PERT CHART

The Program Evaluation and Review Technique (PERT) is used to visualize project activities and dependencies.

Requirement Analysis

↓

System Design

↓

Database Design

↓

Development

↓

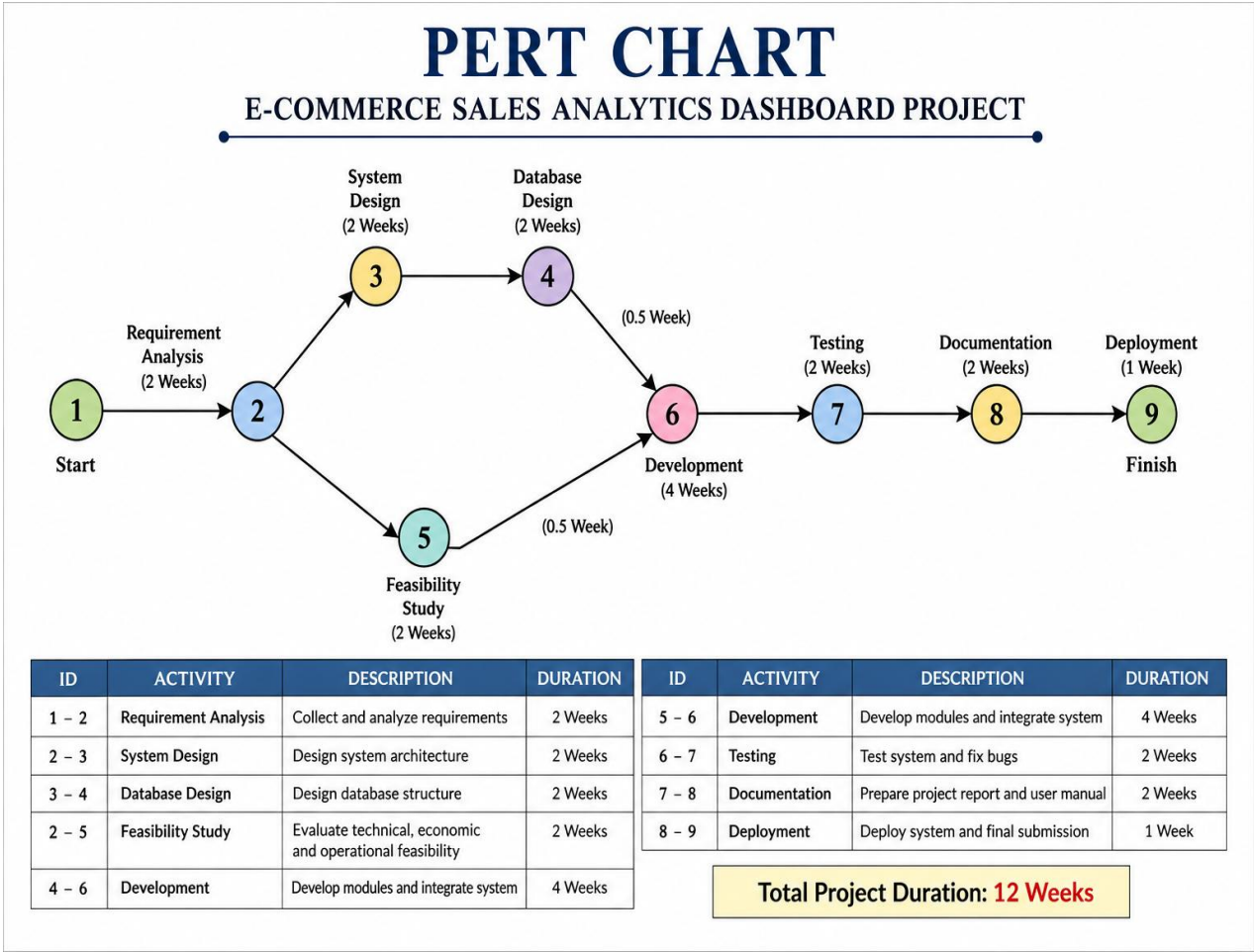
Testing

↓

Documentation

↓

Deployment



CHAPTER CONCLUSION

In this chapter, the E-Commerce Sales Analytics Dashboard was introduced and the objectives, project type, benefits, hardware and software requirements, problem statement, requirements analysis, project plan, Gantt chart, and PERT chart were described.

Chapter two is on System Analysis and Feasibility Study where the need for the system and its technical, economic, and operational feasibilities will be examined.

CHAPTER 2

SYSTEM ANALYSIS

2.1 Introduction

System Analysis is one of the most vital stages of software development. It entails examining the existing system, finding out its weaknesses, determining the needs of the users and designing a better system. In case of E-Commerce Sales Analytics Dashboard, it aids in finding out ways of collecting, analyzing, processing and presenting sales data in a useful manner.

The major objective of this stage is to find out the needs of users such as managers, administrators and analysts who need reports on sales.

E-Commerce Sales Analytics Dashboard is a system that converts sales data into valuable information using charts, graphs, filters and reports.

2.2 Need Identification

In today's business environment, a lot of sales information is generated daily through digital platforms. It is not an easy task to analyze and manage these sales information manually.

Problem with Present System

- Manual analysis of sales information.
- Report generation is a time-consuming process.

- Difficulties in identification of sales trend.
- Non-availability of real-time visualization.
- Wrong decisions because of scattered information.
- Absence of any centralized system for analyzing the information.
- Chances of human error are high.
- Less business insights.
- Requirement for Proposed System

The proposed system should be capable of:

- Analyzing large amounts of sales data.
- Automatic generation of graph-based reports.
- Provision of accurate sales information.
- Reduction of manual efforts.
- Better decision-making.
- Region and category wise analysis.
- User authentication and security.
- Downloading of report.

2.3 Feasibility Study

A feasibility study determines whether the project is practical and beneficial.

The feasibility study is divided into:

1. Technical Feasibility
2. Economic Feasibility
3. Operational Feasibility

2.3.1 Technical Feasibility

Technical feasibility examines whether the required technology, hardware, and software resources are available.

Technologies Used

Component	Technology
Frontend	Streamlit
Backend	Python
Database	SQLite
Data Processing	Pandas
Visualization	Plotly, Matplotlib
Development Tool	VS Code
Deployment	Streamlit Cloud

Technical Advantages

- Open source technologies.
- Easy to deploy.
- Requires very little hardware resources.
- Fast development process.
- Handles large datasets.
- Easy maintenance.

Technical Feasibility Conclusion

The project is technically feasible because all the necessary tools and technologies are easily available and suitable for implementation.

2.3.2 Economic Feasibility

Economic feasibility evaluates the cost-benefit ratio of the project.

Development Cost

Item	Cost
Python	Free
Streamlit	Free
SQLite	Free
VS Code	Free
GitHub	Free
Streamlit Cloud	Free

Advantages

- Decrease in manual effort.
- Increase in speed of reporting.
- Improved decision making.
- Sales performance tracking improved.
- Increased productivity.

Conclusion on Economic Viability

Since the cost of development is negligible because the tools used are free and open-source, it can be concluded that the project is economically viable.

2.3.3 Operational Feasibility

Operation feasibility evaluates the ability of the system to operate well for end-users.

Benefits of Operations

- Friendly user interface.
- Easy login process.
- Interactive dashboard.
- Fast reporting.
- Easy navigation.
- Minimal training needed.
- User Acceptance

This system will be suitable for use by:

- Business Managers
- Sales Analysts
- Administrators
- Company Management

Conclusion on Operational Feasibility

This system is very operationally feasible due to its ease of use and satisfaction of user needs.

2.4 Analysis of Existing System

The current system uses manual techniques like using spreadsheets and conventional reporting.

Workflow of Existing System

Sales Data → Spreadsheets → Analysis → Reporting

Limitations

- Redundant data.
- Manual computations.
- Inefficient processing speed.
- Lack of visualization capability.
- Human error.
- Inefficient data filtering.

2.5 Analysis of Proposed System

The proposed E-commerce Sales Analysis Dashboard automates the process of analysis.

Proposed Workflow

Sales Data Set -> Processing -> Dashboard -> Visualization -> Reports

Features

- Authentication System
- User Authentication
- Administrator Interface
- User Management
- Sales Dashboard
- Key Performance Indicator Cards
- Interactive Graphs
- Filters
- Export to Reports
- Activity Log

Advantages

- Real-time analysis.
- Accurate information.
- Quick reports creation.
- Interactive charts.
- Productivity boost.

2.6 Data Flow Diagram (DFD)

Level 0 DFD

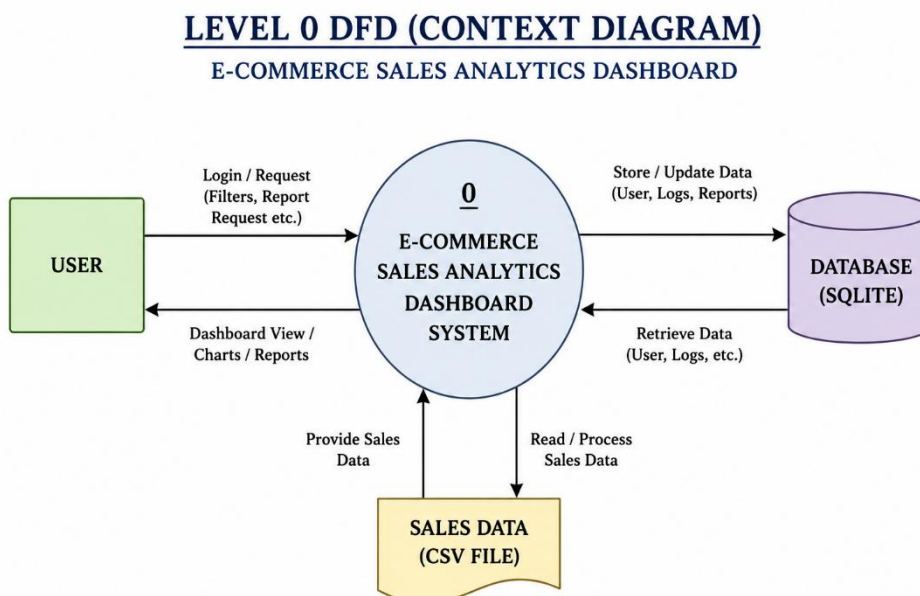


Figure 2.1: Level 0 Data Flow Diagram of E-Commerce Sales Analytics Dashboard

Description

The user interacts with the dashboard. The dashboard processes requests and retrieves data from the database to generate reports and visualizations.

2.7 Modules Description

This application contains the following modules:

Module 1: User Authentication

Features

- Login
- Logout
- Session Management

Purpose

For restricting access to the dashboard for unauthorized users.

Module 2: Sales Data Processing

Features

- Data Cleansing
- Data Transformation
- Data Aggregation

Purpose

For turning raw data into useful information.

Module 3: Dashboard Visualization

Features

- KPI Cards
- Charts
- Graphs
- Sales Trend

Purpose

For displaying business performance data.

Module 4: User Management

Features

- Add User
- View Users
- Manage Roles

Purpose

For managing users.

Module 5: Activity Log

Features

- Login Activities
- Logout Activities
- User Activities

Purpose

For maintaining activity logs.

Module 6: Reporting

Features

- Generate PDF Reports
- Download Reports

Purpose

For generating business reports.

2.8 Data Structure Snapshot

Users Table

Field	Type
Username	Text
Password	Text
Role	Text

Logs Table

Field	Type
Username	Text
Action	Text
Time	DateTime

Sales Data Table

Field	Type
Order ID	Integer
Product	Text
Category	Text
Region	Text
Sales	Float
Profit	Float

2.9 List of Reports

The system provides the following reports:

Sales Reports

- Total Sales Report
- Monthly Sales Report
- Regional Sales Report
- Category Sales Report

Profit Reports

- Profit Analysis Report
- Trend of Profit Report
- Customer Reports
- Customer Insights Report
- Customer Purchase Behaviour Report

User Reports

- User Activity Logs
- Login History Report

2.10 Scope and Future Enhancements

Current Scope

- Sales Analytics
- Visualization in Dashboard Form
- User Authentication
- Report Generation
- Monitoring Activity

Future Enhancements

- Sales Prediction Using AI
- Integration of Machine Learning
- Mobile Application Development
- Database Cloud Hosting
- Data Streaming in Real Time
- Automated Email Report
- Security Enhancements

CHAPTER 3

SYSTEM ANALYSIS

3.1 INTRODUCTION

System Analysis is an investigation of an existing system for defining the requirements of a new system. This allows identification of the problems, analysis of the requirements of the users and designing of efficient systems.

In case of the project on E-Commerce Sales Data Analytics Dashboard, the system analysis is concerned with the analysis of the sales data. The proposed system converts the data of the sales into useful information by means of the use of dashboards and reports.

The main aim of the system analysis is to ensure that the developed system meets the business needs and is reliable, user-friendly and scalable.

3.2 NEED IDENTIFICATION

The e-commerce ventures of today have large volumes of sales data generated on a daily basis. Manually analyzing such data can be both hard and cumbersome.

Problems existing in conventional methods:

- High volume of sales data.
- Hard to track sales trends.
- Manual generation of reports.
- Lack of real-time insights.
- Trouble finding profitable products.
- Bad decision making owing to fragmented data.

Requirement of Proposed System:

- Automatic data analysis.
- Interactive dashboard.
- Fast reporting.
- Good insights into business.
- User-friendly interface.
- Secure login.

3.3 EXISTING SYSTEM

The current system relies heavily on spreadsheets and manual calculations.

Characteristics:

- Information in Excel sheets.
- Manual sales analysis.
- Static reports.
- Lengthy computations.
- Error-prone.
- No dashboard available.

Challenges:

- Slow computation.
- Errors by humans.

- Hard to handle large amounts of data.
- No real-time visualization.
- No advanced filters.

3.4 PROPOSED SYSTEM

The suggested E-Commerce Analytics Dashboard system is an automated tool for sales analysis and visualization.

Key Features:

- Secure Login System
- Role-Based Access
- Sales Dashboard
- Regional Analysis
- Category Analysis
- Customer Insights
- Monthly Trends Analysis
- Report Generation in PDF Format
- User Activity Logs
- Administration of Users

Advantages:

- Fast analysis.
- Accurate reports.
- Better decision-making.
- Better user experience.
- Simple data visualization.

3.5 FEASIBILITY STUDY

Feasibility Study checks the possibility of successful implementation of the project.

3.5.1 Technical Feasibility

Technical feasibility studies the availability of necessary technology.

Tools Used:

- Python
- Streamlit
- SQLite
- Pandas
- Plotly
- ReportLab

Advantages:

- Tools are open source.
- Easier implementation.
- Fewer hardware requirements.
- Maintenance is easier.

Conclusion:

Technical feasibility of the project.

3.5.2 Economic Feasibility

Economic feasibility evaluates project cost.

Software Cost:

Python – Free

Streamlit – Free

SQLite – Free

Pandas – Free

Plotly – Free

Hardware Cost:

Existing computer system can be used.

Conclusion:

The project is economically feasible because development cost is very low.

3.5.3 Operational Feasibility

Operational feasibility determines whether users can operate the system efficiently.

Benefits:

- User-friendly interface.
- Easy navigation.

- Minimal training required.
- Interactive reports.

Conclusion:

The system is operationally feasible.

3.6 REQUIREMENT ANALYSIS

Requirements can be of two types; Functional requirements and non-functional requirements.

3.6.1 Functional Requirements

- User Login
- User Authorization
- Admin Administration
- Display Dashboard
- Analysis of Sales
- Filtering Data
- Generating PDF Reports
- Log Activity
- Logout

3.6.2 Non-Functional Requirements

- Security
- Performance
- Reliability
- Scalability
- Maintainability
- Usability

3.7 INPUT ANALYSIS

Inputs to the system are as follows:

- Username
- Password
- Sales Data Set

- Product Details
- Region Details
- Customer Details
- Category Filters

3.8 PROCESS ANALYSIS

The process involves processing of data in several steps:

Step 1:
Login By User

Step 2:
Verification

Step 3:
Loading of Data Set

Step 4:
Cleaning of Data

Step 5:
Sales Calculation

Step 6:
Generation of Visualization

Step 7:
Report Creation

Step 8:
Result Display

3.9 OUTPUT ANALYSIS

Outputs produced by the system include:

- Key Performance Indicators (KPI) cards
- Sales reports
- Trends charts

- Categorization analysis
- Regional analysis
- Customer analysis
- PDF report
- Logs

3.10 DATA FLOW ANALYSIS

Data flow analysis within the system is as follows:

User → Login Module → Database

Database → Analytics Engine

Analytics Engine → Dashboard

Dashboard → Reports

Reports → User

3.11 MODULE DESCRIPTION

Module 1: Login Module

Functions:

- Authentication
- User Verification
- Access Control

Module 2: Dashboard Module

Functions:

- KPI Presentation
- Charts
- Graphs

Module 3: Analytics Module

Functions:

- Data Analysis
- Trend Analysis
- Sales Analysis

Module 4: Report Module

Functions:

- Report as PDF
- Downloading Reports

Module 5: Admin Module

Functions:

- Creating User
- Viewing Users
- Managing Users

Module 6: Activity Log Module

Functions:

- Login Log
- Logout Log
- Logging User Activities

3.12 SYSTEM BENEFITS

- Simple to use.
- Quick processing.
- Safe authentication.
- Interactive interface.
- Correct analysis.
- Professional reports.
- Improved business decisions.

3.13 SYSTEM DRAWBACKS

- Needs uploaded data set.
- Few offline capabilities.
- Simple user role management.
- Internet needed for implementation.

3.14 POSSIBLE IMPROVEMENTS IN THE FUTURE

- Artificial intelligence predictions.
- Sales forecasts.
- Mobile app.
- Cloud-based database.
- Real time analysis.
- Email generation of reports.
- User roles improvement.
- Integration of APIs.

CHAPTER 4: SYSTEM DESIGN

4.1 Introduction

System Design is the activity of determining the architecture, components, modules, interfaces, and data structures of a system to satisfy certain requirements. The E-Commerce Sales Data Analytics Dashboard aims to design a platform that facilitates efficient sales performance analysis for users.

This step converts the requirements obtained through system analysis into a design that helps in the implementation of the solution. The design consists of database design, module design, interface design, and process design.

4.2 System Architecture

The architecture of the project is based on three layers:

- Presentation Layer (Frontend)
- Streamlit User Interface
- Dashboard Components
- Charts and Filters

- Login and Authentication Pages
- Business Logic Layer
- Data Processing
- User Authentication
- Reporting
- Sales Analysis
- Data Layer
- SQLite Database
- Sales Dataset
- User Details
- Logs

3.3 Input Design

Input design helps in accurate and efficient data entry.

Inputs:

A. Login Details

- Username
- Password

B. User Registration

- Username
- Password
- Role

C. Dashboard Filter

- Filter Region
- Filter Category
- Filter Date

Objectives of Input Design:

- To minimize mistakes
- Improves usability
- Ensures consistency of data

- Security

4.4 Output Design

Output design is about information presentation.

Outputs Produced:

- Total Sales
- Total Profit
- Total Orders
- Graphical Representation of Sales Trend
- Sales Analysis by Category
- Sales Analysis by Region
- User Activity Log
- PDF Report

Benefits:

- Easy to interpret
- Improved decision-making
- Visual representation

4.5 Database Design

Database used by the system is SQLite.

Table: Users

Fields:

- ID
- Username
- Password
- Role

Purpose: Authentication and role information for users.

Table: Logs

Fields:

- ID
- Username
- Action
- Timestamp

Purpose: Records activities performed by the users such as login and logout actions.

4.6 Description of Entity-Relationship (ER) Diagram

Entities:

User

- User ID
- Username
- Password
- Role

Activity Log

- Log ID
- Username
- Action
- Timestamp

Relationship:

User ———-> Log Entry

This relationship helps to keep track of activities performed by each user.

4.7 Module Design

Module 1: Login Module

Function:

- User Authentication
- User Role Verification
- Session Management

Module 2: Dashboard Module**Functions:**

- Key Performance Indicators Display
- Data Filtering
- Charts Display

Module 3: User Management Module**Functions:**

- User Addition
- User Listing
- User Role Management

Module 4: Report Generation Module**Functions:**

- Report Generation
- PDF File Export
- Analysis Download

Module 5: Activity Monitoring Module**Functions:**

- User Login Log
- User Logout Log
- User Activity Log

4.8 Data Flow Explanation

Step 1:

Credentials are entered by the user.

Step 2:

Authorization takes place.

Step 3:

Access is granted.

Step 4:

Sales data set is imported.

Step 5:

Filters are applied by the user.

Step 6:

Data analysis takes place.

Step 7:

Graphs and reports are displayed.

Step 8:

Logs are maintained.

4.9 User Interface Design

Dashboard Elements:

Header Area

- Title of Project
- Navigation

Sidebar

- Filter for Region
- Filter for Category

Dashboard

- KPIs Cards
- Graphs
- Tables

Administrator Panel

- Users Database
- Create User
- Activity Logs

4.10 Security Design

Implemented security measures include:

- Authorization system
- Access control based on roles
- Session management
- Database security
- Input validation

Advantages:

- No unauthorized access
- Security for sensitive information
- Reliability

4.11 Report Design

Reporting is made possible by the dashboard.

Report Content:

- Sales summary
- Analysis of profit
- Category performance
- Regional performance
- Information about user activity

Export Formats:

- PDF
- CSV

4.12 Strengths of the Designed System

- User-Friendly Interface
- Rapid Processing Speed
- Security in Access Control
- Interactive Dashboard
- Real-Time Data Analysis
- Reporting
- Efficient User Management

4.13 Chapter Summary

In this chapter, the entire design of the E-Commerce Sales Data Analytics Dashboard was explained. The architectural design, database design, module design, user interface design, security system, and reporting process were covered. This design serves as the basis for the development of a secure and efficient analytics dashboard.

CHAPTER 5

IMPLEMENTATION METHODOLOGY

5.1 INTRODUCTION

Implementation refers to the process through which the proposed system becomes a functional application. During this process, all the designs, requirements, and planning that were done in the earlier stages of the project will be turned into functional parts of the application.

The implementation of the E-Commerce Sales Data Analytics Dashboard was done using the following technologies: Python, Streamlit, SQLite, Pandas, Plotly, and ReportLab. This process included the creation of the login, dashboard, user management, database, logs, and reporting systems.

The main aim of implementation is to make sure that all the operations of the system are completed effectively and accurately.

5.2 ENVIRONMENT OF IMPLEMENTATION

Environment of implementation used for the project is as follows:

- Programming language:
Python 3.12
- Front-end framework:
Streamlit
- Database:
SQLite
- Data processing:
Pandas
- Data visualization:
Plotly
- Report generation:
ReportLab
- IDE:
Visual Studio Code
- Version control:
Git and GitHub
- Operating system:
Windows 11

5.3 STRUCTURE OF THE PROJECT

The project comprises of multiple files/modules.

Files/Modules of the Project:

- app.py

The main application file.

- database.py

Functions to interact with the database.

- logger.py

Functions for logging the activities.

- sales_data.csv

The dataset containing the sales data.

- database.db

The SQLite database.

- requirements.txt

Dependencies of the project.

- README.md

Documentation of the project.

- sales_report.pdf

The generated report file.

The modular structure improves maintainability and scalability.

5.4 DATABASE IMPLEMENTATION

An SQLite database is employed for the storage of user data and logging.

Users Table

Objective:

Login details and roles of users.

Fields:

Username

Password

Role

Logs Table

Objective:

User actions logs.

Fields:

Username
Action
Timestamp

Strengths:

- Small size
- Quick
- Easy integration
- Server-less
- Ideal for smaller projects

5.5 USER AUTHENTICATION IMPLEMENTATION

Authentication is used to validate user's identity before providing access.

Steps:

Step 1:

Username and password entered by the user.

Step 2:

Database lookup performed.

Step 3:

Validation of credentials performed.

Step 4:

Identification of the role.

Step 5:

Access provided.

Advantages:

- Ensures secure access
- Increases security
- Sensitive data protection

5.6 LOGIN MODULE IMPLEMENTATION

The login module serves as an entry point of the application.

Components:

- Validation of username
- Validation of password
- Verification of role
- Session handling

Functions:

Login

Logout

Session Management

Advantages:

- Secured access
- Enhanced user experience
- Controlled authentication

5.7 SESSION MANAGEMENT

Session state is managed by the Streamlit Session State.

Functions:

- Keeping login status
- Saving role of the user
- Saving username
- Restricting unauthorized access

Advantages:

- Improved security
- Enhanced navigation
- Improved user experience

5.8 USER MANAGEMENT IMPLEMENTATION

Admin user has access rights to create users.

Features:

- Create User
- Assign Role
- Display Users Database

Validation Implementation:

Username should not be blank.

Password should be at least six characters long.

There should be no duplicate usernames.

Role should be chosen.

Advantages:

- Enhanced data integrity
- Increased security
- Simplified management

5.9 DATA PROCESSING IMPLEMENTATION

Sales dataset is analyzed using Pandas.

Processes Conducted:

- Data Importation
- Data Cleaning
- Data Filtering
- Data Aggregation
- Statistical Analysis

Advantages:

- Speedy processing
- Effective analysis
- Precise results

5.10 IMPLEMENTATION OF DASHBOARD

The dashboard is the key element of the project.

Functions:

- Showing KPIs
- Displaying graphs
- Filtering data
- Deriving insights

Sections of the Dashboard:

A. KPI cards

Shows:

- Total Sales
- Total Profit
- Total Orders
- Total Customers

B. Filters Section

Enables users to choose:

- Region
- Category

C. Visualizations Section

Show graphs and charts.

Advantages:

- Interactive
- Easily understandable
- Rapid analysis

5.11 DATA VISUALIZATION IMPLEMENTATION

Data visualization implementation is done using Plotly.

Implemented Charts:

Sales Trends Chart

Usage:

To visualize sales growth with time.

Category-wise Sales Chart

Usage:

To visualize category wise performance.

Region-wise Sales Chart

Usage:

To visualize sales in different regions.

Profit Analysis Chart

Usage:

For visualization of profit trends.

Advantages:

- Interactive
- Visually attractive
- Easy to interpret

5.12 ADMIN PANEL IMPLEMENTATION

Admin panel can be accessed by administrators only.

Functions:

- Create User

- User database
- User activities monitoring

Advantages:

- Centralized control
- User management
- Monitoring activities

5.13 ACTIVITY LOGGING IMPLEMENTATION

User activities are stored in the log table.

Logged Activities:

- Login
- Logout

Advantages:

- Audit trail
- Monitoring security issues
- Accountability

Details logged in the log table:

- Username
- Action
- TimeStamp

5.14 PDF REPORT GENERATION

ReportLab library is used to generate PDF reports

.

Report Contents:

- Sales Summary
- Profit Analysis
- Dashboard Metrics
- User Information

Benefits:

- Professional reports
- Easy sharing
- Offline availability

5.15 IMPLEMENTATION OF ERROR HANDLING

The system is capable of handling most errors appropriately.

Examples:

- Invalid Login Attempt
- Invalid Username
- Duplicated Username
- Database Errors

Advantages:

- Improved user experience
- System stability
- Reliability improvement

5.16 IMPLEMENTATION OF SECURITY

Security features implemented include:

- User Authentication
- User Authorization
- User Session Management
- Role-based access control
- Activity logs

Advantages:

- Prevention of misuse
- Data protection
- Improved reliability

5.17 TESTING IN IMPLEMENTATION PHASE

Some tests were conducted during implementation phase.

- Unit Test:
Test individual modules.
- Integration Test:
Test modules interaction.
- System Test:
Test entire application.
- User Acceptance Test:
Test usability and functionality.

Conclusion:

All modules worked properly.

5.18 PROBLEMS ENCOUNTERED DURING IMPLEMENTATION

- Database Integration Problems.
- Session Management Problems.
- User Authentication Problems.
- Report Generation Issues.
- Duplicate Users.

Solutions:

- Database redesigning
- Implementing validation
- Improved error handling
- Testing and debugging

5.19 ACHIEVEMENTS OF IMPLEMENTATION

The implemented system has been able to:

- Provide login securely
- Have interactive dashboard
- Manage users
- Log activities
- Analyze data
- Generate reports in PDF

5.20 CHAPTER CONCLUSION

In this chapter, the full implementation approach for E-Commerce Sales Data Analytics Dashboard was highlighted. The development environment, database implementation, authentication system, dashboard modules, user management, activity logging, report generation, and security were all described. Through this implementation, the designed system has been converted to an analytics application that can provide sales insights.

CHAPTER 6

SOFTWARE TESTING

6.1 INTRODUCTION

Software Testing is an essential component of the Software Development Life Cycle (SDLC). This entails testing a piece of software or application to detect errors, defects, or bugs prior to its implementation. This phase guarantees that the software is meeting user needs and performing properly under various conditions.

The E-Commerce Sales Data Analytics Dashboard was fully tested to make sure that all the modules perform effectively. The testing process makes software better in terms of quality, reliability, performance, and security.

Testing ensures that the developed system performs well as per the given specifications and provides accurate output.

6.2 TEST OBJECTIVES

Objectives of software testing include:

- To detect software bugs.
- To check system performance.
- To test user requirements.
- To enhance reliability.
- To maintain accuracy.
- To test security.
- To increase customer satisfaction.
- To reduce maintenance cost.
- To check system performance.

6.3 NECESSITY OF TESTING

Testing becomes inevitable since there could be some errors in software which might influence performance and functionality of the system.

Why Testing Is Needed:

- Detection of coding errors.
- To check calculation.
- To improve quality of the software.
- To check security.
- To check output.
- To build users' confidence.
- To avoid system failure.

6.4 TESTING TYPES

The following types of testing have been performed in the project:

- Unit Testing
- Integration Testing
- System Testing
- Validation Testing
- Security Testing
- User Acceptance Testing

6.5 UNIT TESTING

Unit testing is the process of testing each module independently.

Modules Test:

- Login Module
- Dashboard Module
- User Management Module
- Activity Logging Module
- Report Generation Module

Advantages:

- Early detection of bugs.
- Debugging is easier.
- Better coding.

Example:

Testing of Login Module

Input:

Username = admin

Password = admin123

Expected Output:

Login Success

Actual Output:

Login Success

Status:

PASS

6.6 INTEGRATION TESTING

Integration testing ensures successful communication between modules.

Modules Under Integration:

Login Module <--> Database

Dashboard Module <--> Dataset

User Management <--> Database

Report Generation <--> Dashboard

Objectives:

- Data flow verification.
- Interaction of modules.
- Issues with integration.

Result:

All modules under integration performed successfully.

6.7 SYSTEM TESTING

The system is tested as a whole for the entire application.

Tested Components:

- User Authentication
- Dashboard Analytics
- Activity Logging
- Report Generation
- User Management

Observation:

The entire system worked as per requirements.

6.8 VALIDATION TESTING

Validation testing validates that the system meets the user requirements.

Validated Requirements:

- Login feature
- Dashboard feature
- User creation
- PDF report generation
- Activity tracking

Conclusion:

All the functional requirements were successfully validated.

6.9 SECURITY TESTING

Security testing ensures protection from any kind of unauthorized access.

Features of Security Tested for:

- Authentication
- Validation of password
- Access Control based on roles
- Sessions

Results of Testing:

There was no unauthorized access observed during testing.

6.10 USER ACCEPTANCE TESTING (UAT)

User Acceptance Testing is done by end-users to ensure that the software fulfills business requirements.

Users Participated in Testing:

- Administrator
- Business user

Feedback Given:

- Very user-friendly.
- Good looks.
- Helpful analytics.
- Quick report preparation.

Result:

Software accepted by users.

6.11 TEST CASES

TEST CASE 1

Test Name:

Login Authentication

Input:

Valid Username and Password

Expected Output:

Login Successful

Actual Output:

Login Successful

Status:

PASS

TEST CASE 2

Test Name:

Invalid Login

Input:

Wrong Password

Expected Output:

Login Failed

Actual Output:

Login Failed

Status:

PASS

TEST CASE 3

Test Name:

Create New User

Input:

Username = Noor

Password = 123456

Role = User

Expected Output:

User Created Successfully

Actual Output:

User Created Successfully

Status:

PASS

TEST CASE 4

Test Name:

Duplicate Username

Input:

Existing Username

Expected Output:

Username Already Exists

Actual Output:

Username Already Exists

Status:

PASS

TEST CASE 5

Test Name:

Empty Username

Input:

Blank Username

Expected Output:

Username Cannot Be Empty

Actual Output:

Username Cannot Be Empty

Status:

PASS

TEST CASE 6

Test Name:

Password Validation

Input:

123

Expected Output:

Password Must Be At Least 6 Characters

Actual Output:

Password Must Be At Least 6 Characters

Status:

PASS

TEST CASE 7

Test Name:

Dashboard Loading

Input:

Login Successful

Expected Output:

Dashboard Displayed

Actual Output:

Dashboard Displayed

Status:

PASS

TEST CASE 8

Test Name:

PDF Report Generation

Input:

Generate Report Button

Expected Output:

PDF Downloaded

Actual Output:

PDF Downloaded

Status:

PASS

TEST CASE 9

Test Name:

Activity Logging

Input:

User Login
Expected Output:
Log Entry Created
Actual Output:
Log Entry Created
Status:
PASS

TEST CASE 10

Test Name:
User Logout
Input:
Logout Button
Expected Output:
Session Closed
Actual Output:
Session Closed
Status:
PASS

6.12 TEST RESULTS SUMMARY

Test Module	Status
Login Module	PASS
Dashboard Module	PASS
User Management	PASS
Activity Logging	PASS
PDF Report Generation	PASS
Database Operations	PASS
Validation Mechanism	PASS
Security Features	PASS

Overall System Status:
SUCCESSFUL

6.13 ERRORS IDENTIFIED DURING TESTING

The errors that were discovered during development include the following:

Session State Error
SQLite Table Not Found Error
Duplicate User Records
PDF Report Formatting Error
User Validation Error
Python Indentation Error

6.14 TAKEN CORRECTIVE ACTIONS

Problem:
Session State Error

Fix:
Correctly initialized session variables.

Problem:
Duplicate User Records

Fix:
Enforced unique username.

Problem:
Weak Passwords

Fix:
Minimum six characters requirement.

Problem:
Database Errors

Fix:
Created tables before executing queries.

Problem:
PDF Formatting Issue

Fix:

Fixed ReportLab formatting.

6.15 PERFORMANCE TESTING

Performance testing was carried out in order to test the performance of the system.

Tested Parameters:

- Login Response Time
- Dashboard Loading Time
- Report Response Time
- Database Querying Performance

Results:

Login Response Time:

Less than 2 seconds

Dashboard Loading Time:

Less than 3 seconds

Report Generation Time:

Less than 5 seconds

Database Queries:

Very fast

6.16 ADVANTAGES OF TESTING

- Higher reliability.
- Higher security.
- Lower bugs.
- Higher user satisfaction.
- Better system performance.
- Higher maintainability.

6.17 LIMITATIONS OF TESTING

- Testing does not ensure the production of error-free software.
- Limited test data.
- Future changes to the system may result in new errors.

6.18 CHAPTER CONCLUSION

This chapter described testing methods utilized for testing the E-Commerce Sales Data Analytics Dashboard. Different kinds of tests such as Unit testing, Integration testing, System testing, Security testing, Validation testing, and User acceptance testing were conducted. It was concluded that all the requirements of the system have been met through testing. All important modules of the system worked properly.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 INTRODUCTION

This chapter discusses the findings that have been realized using the E-Commerce Sales Data Analytics Dashboard. In this chapter, the reader will see how the developed application works in practice, how it can help to analyze the sales data, and how it validates the implementation of the project objectives.

The sales datasets are processed, calculations are done, visualizations are provided, and business insights are generated. These insights help managers to understand the performance of their businesses and plan their future activities accordingly.

The main objective of this chapter is to analyze the performance of the developed system and discuss the results produced by different modules of the system.

7.2 THE PROCESS OF EXECUTING THE SYSTEM

The process of executing the system starts with the user authentication. After successful authentication, the user sees the dashboard with analytical information presented via graphs, charts, KPI cards, and reports.

Execution Flow:

User Login



Authentication Verification



Dashboard Loading



Dataset Processing



Sales Analysis



Visualization Generation



Report Generation



User Logout

The system successfully executes all operations and produces accurate analytical results.

7.3 LOGIN MODULE RESULT

The Login Module performs the function of authentication before providing access to the system.

Purpose of Login Module:

- To ensure that no unauthorized access is made.
- To protect business information.
- To secure the system.
- To know about the role of the user.

Inputs Needed:

Username

Password

Outputs Produced:

Access Granted (User Valid)

Access Denied (User Invalid)

Features Implemented:

- Username Validation
- Password Validation
- Role Validation
- Session Management

Validation Process:

- Username cannot be blank.
- Password cannot be blank.

- Invalid login is not allowed.
- Session information is stored safely.

Advantages:

- More security.
- Improved access control.
- Avoiding unauthorized access.

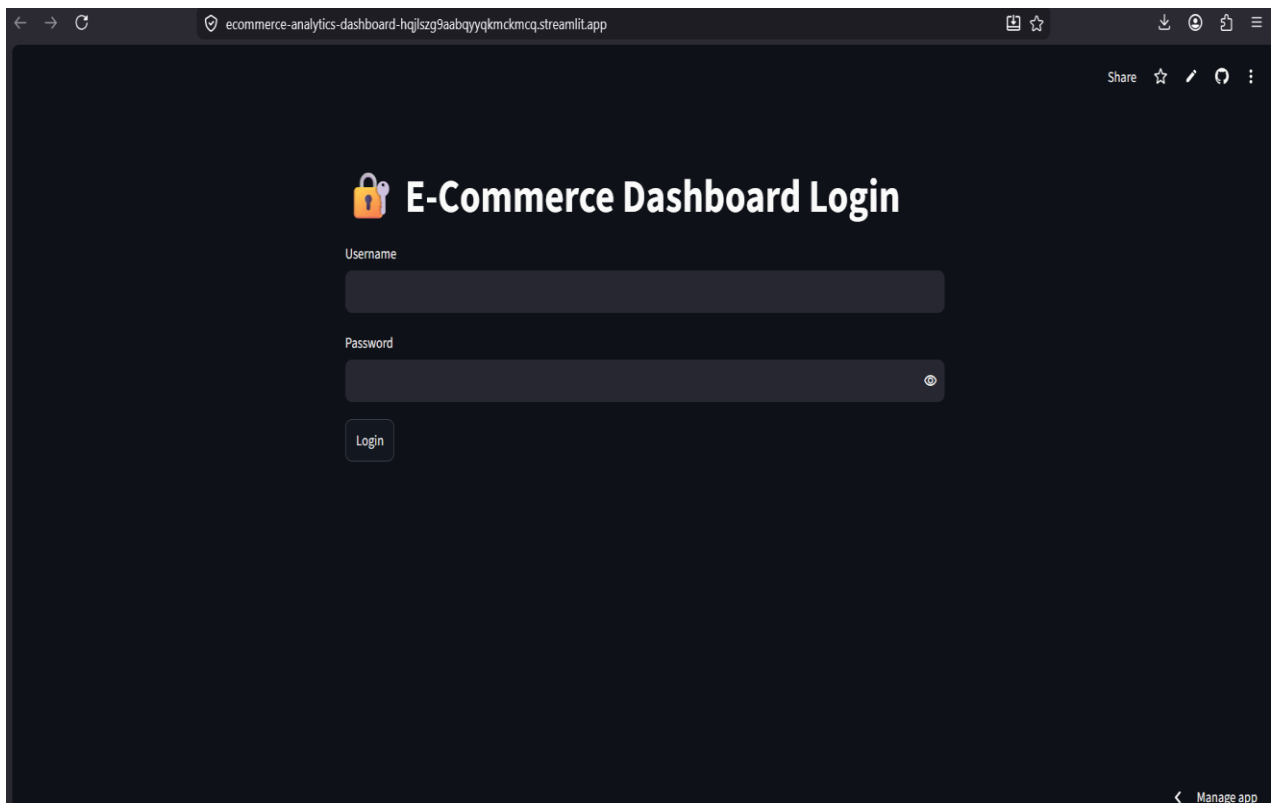


Figure - Login Interface

Discussion:

The Login Module successfully verifies user credentials and ensures that only authorized users can access the dashboard. This improves system security and protects business information from unauthorized access.

7.4 DASHBOARD HOME PAGE RESULT

The Dashboard Home Page acts as the home page for the software. It helps to display all information related to the business by the use of KPI cards and other reports.

Information Shown:

- Total Sales
- Total Profit
- Total Orders
- Sales Trends
- Category Performance
- Regional Performance

Goals:

- Gain insight into the business.
- Cut down on the analysis manually.
- Decision making.

Features:

- Interactive Interface
- The user can interact with the charts and filters provided.
- Centralization of Information
- All necessary information is shown on one page.
- Instant Analysis
- Information is analyzed and then shown.

Benefits:

- Increased productivity.
- Proper visualization.
- Easy navigation.

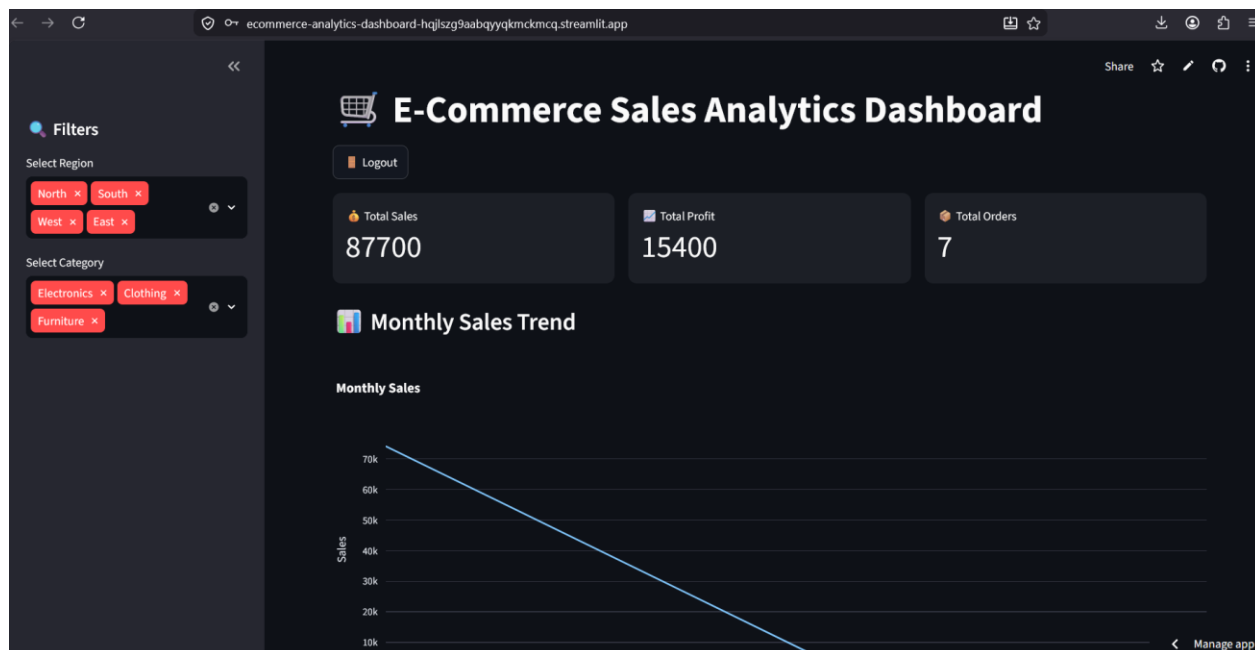


Figure - Dashboard Home Page

Discussion:

The dashboard provides a comprehensive overview of sales performance. Users can quickly understand business conditions without manually reviewing raw datasets.

7.5 KPI CARD ANALYSIS

KPI (Key Performance Indicator) cards display important business metrics.

KPIs Displayed:

- 1.Total Sales
- 2.Total Profit
- 3.Total Orders
- 4.Total Customers

Purpose:

- To make data available instantly.
- To enable faster decision-making.
- To enhance managerial efficiency.

How it works:

The tool computes the value from the data and presents it using KPI cards.

Advantages:

- Ease of interpretation.
- Quicker monitoring.
- Time-saving analysis.

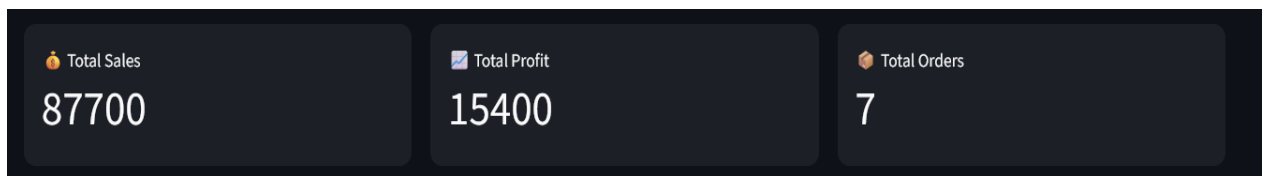


Figure - KPI Cards

Discussion:

The KPI cards successfully provide summarized business information. Users can evaluate organizational performance at a glance.

7.6 SALES TREND ANALYSIS

Sales Trend Analysis can assist in determining sales performance trends.

Purpose:

- Sales growth monitoring.
- Detection of sales seasonality.
- Performance variation detection.

Procedure:

Data loading.

Date grouping.

Sales sums calculation.

Trend chart generation.

Advantages:

- Improved forecasting.
- Business planning.
- Performance appraisal.

Importance for Business:

Sales trends allow organizations to estimate their customers' demands and improve their business processes.

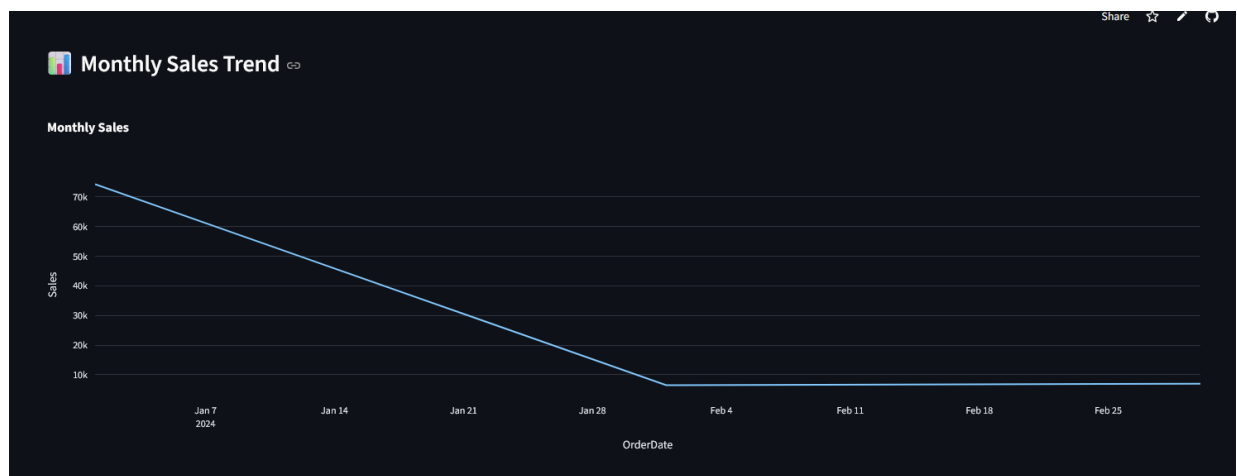


Figure - Sales Trend Analysis

Discussion:

The Sales Trend Chart effectively visualizes changes in sales performance and assists management in identifying growth opportunities.

7.7 ANALYSIS OF SALES ACCORDING TO CATEGORIES

Category-wise analysis is conducted to determine the sales performance within different categories of products.

Objectives:

- Determine best-selling categories.
- Compare performance of categories.
- Optimize inventory management.

Categories:

- Electronics
- Apparel
- Furniture
- Accessories

Advantages:

- Proper inventory management.
- Effective marketing techniques.
- Profitability.

Working Principle:

Sales data is segregated according to the categories and then added up to determine revenue from each category.

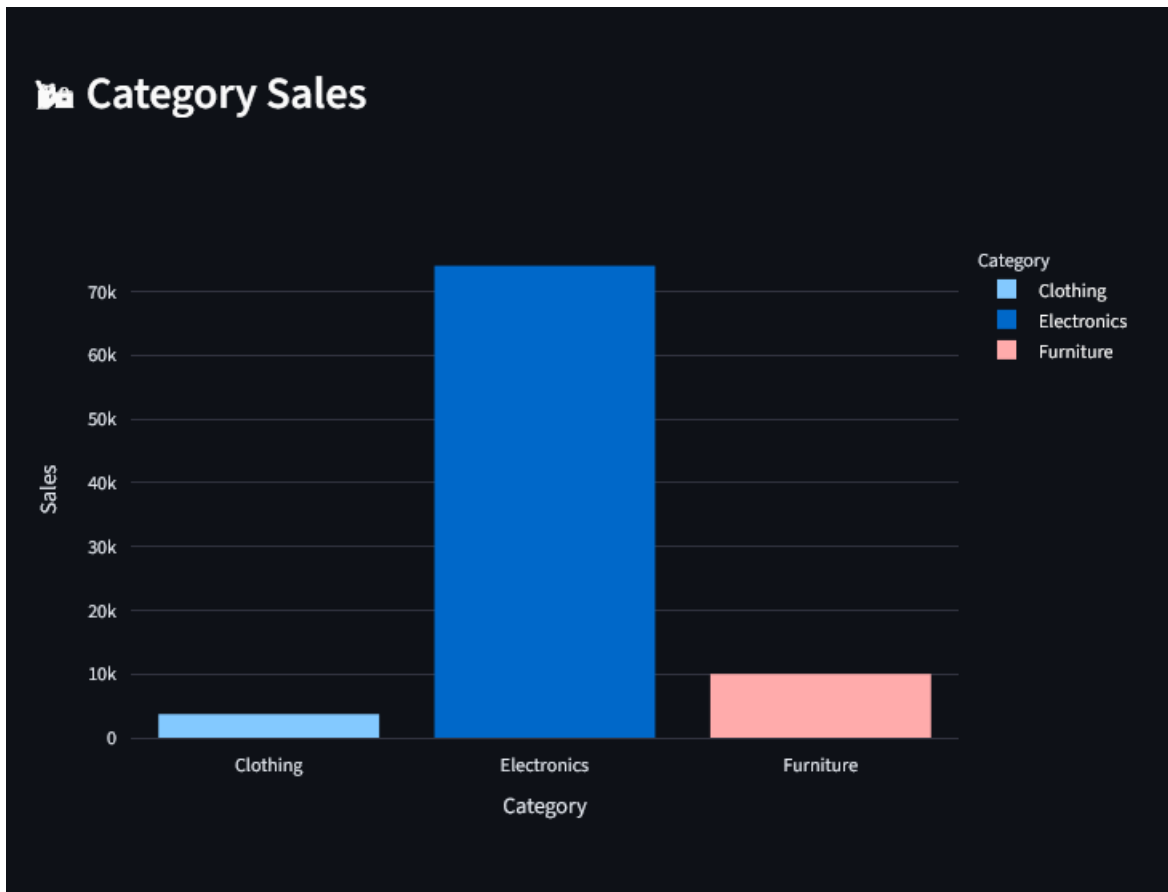


Figure - Category-wise Sales Analysis

Discussion:

The analysis helps management understand which product categories contribute most significantly to business revenue.

7.8 REGIONAL ANALYSIS OF SALES

A region-based analysis focuses on sales performance in various geographical locations.

Goals:

- Compare performances among regions.
- Determine good performing regions.
- Enhance resource management.

Advantages:

- Increased knowledge of the market.
- Effective expansion planning.
- Better distribution management.

Regions Covered:

- North
- South
- East
- West

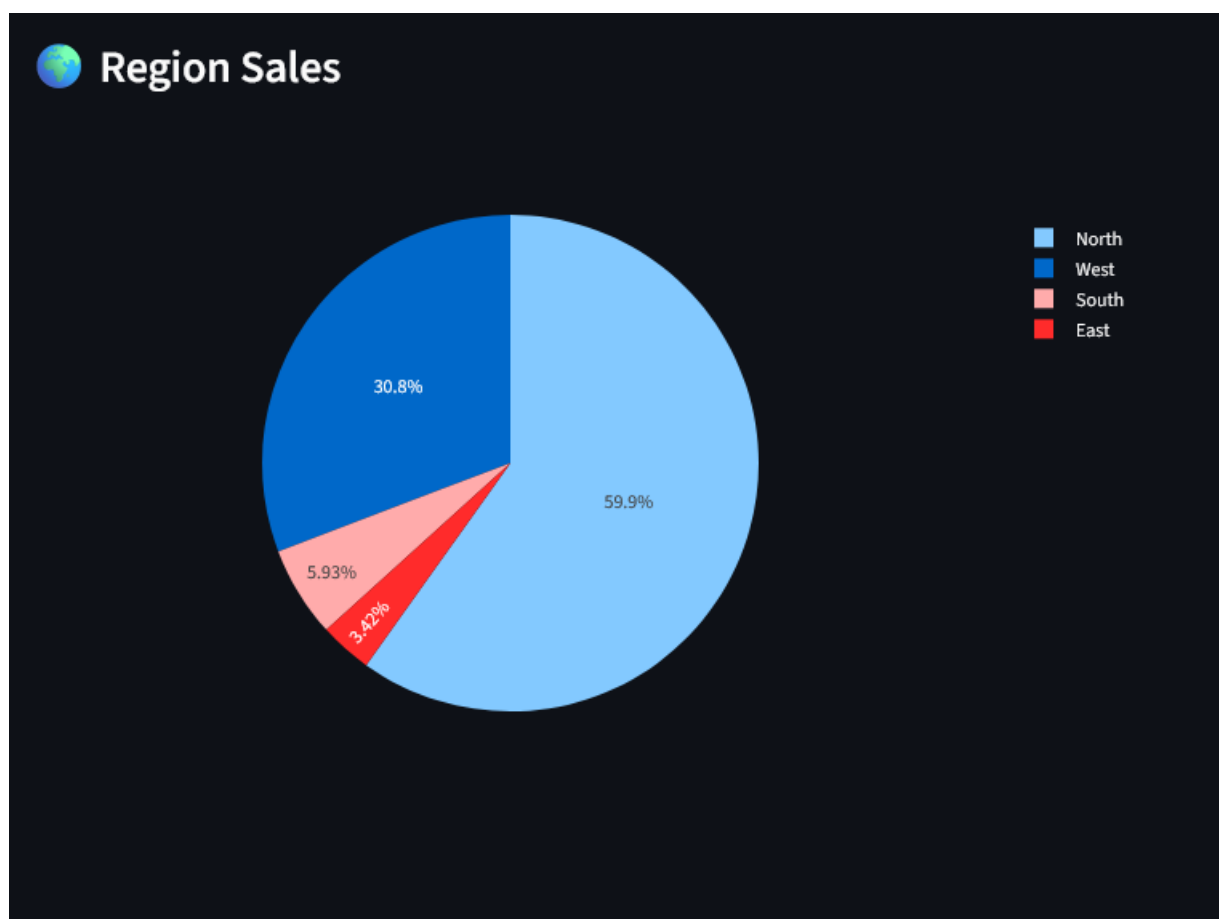


Figure - Region-wise Sales Analysis

Discussion:

Regional analysis provides valuable information regarding market performance and customer demand in different locations.

7.9 PROGRESS ANALYSIS RESULT

The progress analysis measures how successful the company has been financially.

Purpose:

- To measure profitability.
- To assess growth.
- For financial planning.

Displayed Information:

- Total Profit
- Profit Trend Analysis
- Profit Comparison

Advantages:

- Better financial decision-making.
- Cost control.
- Revenue maximization.

Discussion:

Profit analysis assists management in understanding business performance and identifying areas for improvement.

7.10 USER MANAGEMENT RESULT

The User Management Module allows the administrator to manage users in the system.

Functions:

- Add User
- List Users
- Allocate Roles

Validation:

- Username should not be blank.
- Password should be more than six characters long.
- Cannot use duplicate usernames.

Advantages:

- Increased security.
- Efficient management.
- User control.

Discussion:

The module successfully manages user accounts and prevents invalid data entry through validation mechanisms.

7.11 ADMIN PANEL RESULT

The Admin Panel provides administrative control over system operations.

Features:

- User Management
- Activity Monitoring
- Database Access

Advantages:

- Centralized administration.
- Better control.
- Enhanced security.

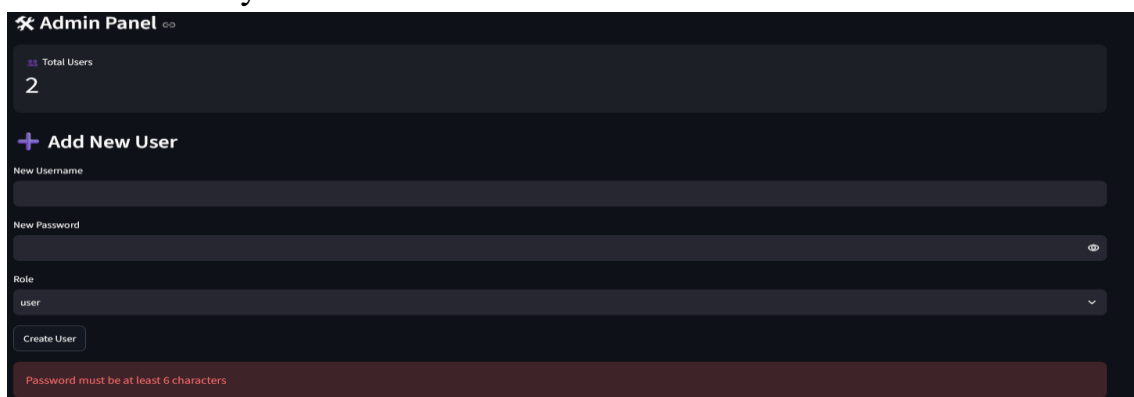
The image shows a dark-themed user interface for an 'Admin Panel'. At the top left, there's a star icon and the text 'Admin Panel' with a double-circle icon. Below this, a section titled 'Total Users' shows the number '2'. Further down, there's a '+ Add New User' button. Below the button are three input fields: 'New Username', 'New Password', and 'Role'. The 'Role' field is a dropdown menu currently showing 'user'. At the bottom left of the form is a 'Create User' button. A red error message at the bottom states 'Password must be at least 6 characters'.

Figure - Admin Panel

Discussion:
The Admin Panel simplifies administrative tasks and improves system management efficiency.

7.12 ACTIVITY LOGGING RESULT

The Activity Logging Module logs all the activities carried out by users in the system.

Activities Logged:

- Login
- Logout

Data Stored:

- Username
- Action
- Date/Time

Advantages:

- Audit trail.
- Security monitoring.
- Accountability of users.

User Activity Logs			
	username	action	time
0	admin	Login	2026-07-16 18:17:09.995491
1	admin	Login	2026-07-16 18:17:48.328786

Figure - Activity Logging

Discussion:
Activity logging improves transparency and allows administrators to monitor user behavior effectively.

7.13 PDF REPORT GENERATION RESULT

The dashboard provides downloadable PDF reports.

Report Contents:

- Total Sales
- Total Profit
- Total Orders
- Report Date
- Executive Summary

Advantages:

- Easy sharing.
- Professional documentation.
- Offline access.

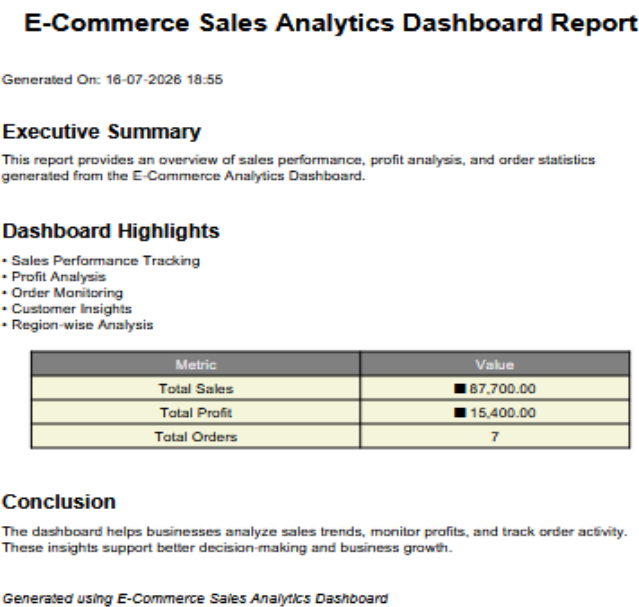


Figure - PDF Report Generation

Discussion:

The PDF reporting feature enables users to generate professional business reports quickly and efficiently.

7.14 DATABASE RESULT

The SQLite database stores all application data.

Database Components:

Users Table

Logs Table

Users Table Fields:

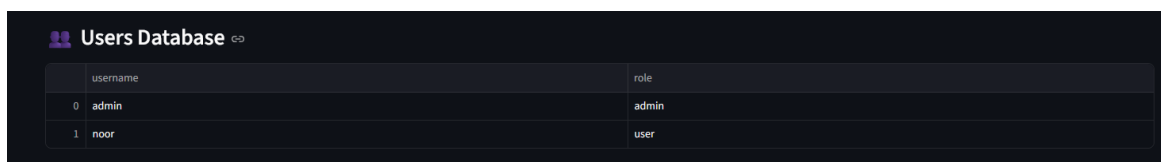
- Username
- Password
- Role

Logs Table Fields:

- Username
- Action
- Timestamp

Advantages:

- Lightweight
- Fast
- Secure
- Easy Integration



	username	role
0	admin	admin
1	noor	user

Figure - Users Table

Discussion:

The database successfully stores and retrieves information required by the application.

7.15 PERFORMANCE TESTING

Performance testing was carried out to measure responsiveness and efficiency.

Tests Carried Out:

Login Time

Time taken to load the Dashboard

Time taken to generate reports

Query Execution Speed

Results:

Login Time:

Less Than 2 seconds

Dashboard Loading Time:

Less Than 3 seconds

PDF File Generation Time:

Less than 5 seconds

Query execution speed:

Efficient

Discussion:

The system performs well and fulfills the expected requirements.

7.16 USER FEEDBACK ANALYSIS

Feedback was collected from users who interacted with the dashboard.

User Type	Feedback
Student	Interface is user-friendly
Faculty	Reports are informative
Administrator	User management is easy

Analysis:

Most users reported positive experiences and found the dashboard useful for analytical tasks.

7.17 COMPARISON OF EXISTING AND PROPOSED SYSTEM

Feature	Existing System	Proposed System
Data Analysis	Manual	Automated
Reports	Static	Dynamic
Security	Limited	Authentication Based
Visualization	Minimal	Interactive
User Management	Not Available	Available
Report Generation	Manual	Automatic

Discussion:
The proposed system significantly improves functionality, efficiency, and usability compared to traditional approaches.

7.18 ACHIEVEMENTS OF THE PROJECT

The project successfully achieved:

- Secure Authentication
- Role-Based Access Control
- User Validation
- Interactive Dashboard
- Data Visualization
- User Management
- Activity Logging
- PDF Report Generation
- Database Integration
- Business Analytics

7.19 OVERALL SYSTEM EVALUATION

The E-Commerce Sales Data Analytics Dashboard successfully meets all functional and non-functional requirements identified during system analysis.

Major Achievements:

- Accurate sales analysis.
- Professional reporting.
- Better decision support.
- Secure access control.
- Efficient data management.

The system demonstrates reliability, usability, and scalability, making it suitable for practical business applications.

7.20 CHAPTER SUMMARY

This chapter presented detailed results generated by the E-Commerce Sales Data Analytics Dashboard. The outputs demonstrate that the system successfully analyzes sales data, generates meaningful insights, provides professional reports, and supports business decision-making. The results confirm that all project objectives have been achieved successfully and the system performs efficiently in a practical environment.

CHAPTER 8

SECURITY MECHANISM

8.1 INTRODUCTION

Security is one of the most important aspects of any software system. In today's digital environment, organizations store large amounts of valuable data, making information security a critical requirement. A secure system protects sensitive information from unauthorized access, modification, disclosure, or destruction. The E-Commerce Sales Data Analytics Dashboard contains important business information such as sales records, user accounts, reports, and activity logs. Therefore, implementing appropriate security measures is necessary to maintain data integrity, confidentiality, and availability.

This chapter discusses the various security mechanisms implemented in the project to ensure secure access and protect organizational data.

8.2 OBJECTIVES OF SECURITY MECHANISM

The primary objectives of implementing security mechanisms are:

- Protect sensitive business information.
- Prevent unauthorized access.
- Maintain data integrity.
- Ensure confidentiality of user credentials.
- Monitor user activities.
- Improve system reliability.
- Reduce security risks.
- Enhance user trust.

Security helps create a safe environment where users can access information without fear of data loss or misuse.

8.3 IMPORTANCE OF SECURITY IN E-COMMERCE ANALYTICS

E-commerce organizations handle large amounts of business data every day. This information is valuable and must be protected against security threats.

Importance of Security:

1. Protection of Business Data

Sales information is a valuable business asset. Security mechanisms ensure that data remains protected.

2. Prevention of Unauthorized Access

Only authorized users should be able to access the dashboard.

3. Protection Against Data Manipulation

Security prevents unauthorized users from modifying analytical data.

4. Improved User Confidence

Users trust systems that implement proper security measures.

5. Compliance with Organizational Policies

Secure systems help organizations follow security standards and best practices.

8.4 AUTHENTICATION MECHANISM

Authentication is the process of verifying the identity of a user before granting access to the system.

In the E-Commerce Sales Analytics Dashboard, authentication is implemented through a username and password mechanism.

Authentication Process:

Step 1:

User enters username and password.

Step 2:

System verifies credentials from the database.

Step 3:

If credentials match, access is granted.

Step 4:

If credentials are incorrect, access is denied.

Benefits of Authentication:

- Prevents unauthorized access.
- Protects sensitive information.
- Improves accountability.
- Enhances security.

Authentication is the first layer of defense in the application.

8.5 AUTHORIZATION MECHANISM

Authorization determines what actions a user can perform after successful authentication.

The system uses authorization to control access to specific features.

Purpose of Authorization:

- Restrict access.
- Control permissions.
- Improve security.
- Prevent misuse.

Example:

Administrator:

Can create users and manage system operations.

Normal User:

Can view dashboard reports but cannot manage users.

Benefits:

- Better control.
- Reduced security risks.
- Organized system management.

8.6 ROLE-BASED ACCESS CONTROL (RBAC)

Role-Based Access Control is a security mechanism where permissions are assigned according to user roles.

Roles Implemented:

1.Administrator

Permissions:

- Login
- View Dashboard
- Create Users
- View Logs
- Manage System

2.User

Permissions:

- Login
- View Dashboard

- Analyze Reports

Advantages of RBAC:

- Easy management.
- Improved security.
- Better control over resources.
- Reduced unauthorized access.

Role-based access control ensures that users can only perform tasks relevant to their responsibilities.

8.7 PASSWORD SECURITY

Passwords are one of the most common methods used for authentication. The project implements password validation to improve security.

Password Rules:

- Password cannot be empty.
- Password must contain at least six characters.

Importance of Password Security:

- Protects user accounts.
- Prevents unauthorized access.
- Improves overall system security.

Best Practices:

- Use strong passwords.
- Avoid predictable passwords.
- Change passwords periodically.
- Do not share passwords.

8.8 INPUT VALIDATION SECURITY

Input validation is implemented to prevent incorrect or malicious data from entering the system.

Validation Rules Implemented:

Username Validation:

- Cannot be empty.
- Duplicate usernames are not allowed.

Password Validation:

- Minimum six characters required.

Role Validation:

- User role must be selected.

Benefits:

- Improved data quality.
- Reduced system errors.
- Better security.

Input validation acts as a protective barrier against invalid data.

8.9 DATABASE SECURITY

The SQLite database stores user information and activity logs.

Database Security Measures:

1. Controlled Access

Only authorized application components can access the database.

2. User Authentication

Database information is accessible only after successful login.

3. Data Integrity

Validation ensures accurate data storage.

4. Restricted Operations

Only administrators can perform user management activities.

Advantages:

- Protects stored information.
- Prevents accidental modification.
- Maintains consistency.

8.10 SESSION MANAGEMENT SECURITY

Session management is used to maintain user state after login.

Purpose:

- Keep users authenticated.
- Track user activity.
- Manage permissions.

Implementation:

The project uses Streamlit Session State to store:

- Username
- User Role
- Login Status

Benefits:

- Better security.
- Improved user experience.
- Controlled access.

Session management prevents unauthorized users from accessing protected pages.

8.11 ACTIVITY LOGGING SECURITY

Activity logging records actions performed by users.

Activities Recorded:

- Login
- Logout

Information Stored:

- Username
- Action
- Timestamp

Advantages:

- Audit trail.
- Accountability.
- Security monitoring.

Importance:

Activity logs help administrators identify suspicious activities and maintain records of user actions.

8.12 DATA CONFIDENTIALITY

Confidentiality ensures that information is accessible only to authorized users.

Methods Used:

- Authentication
- Authorization
- Role-Based Access Control

Benefits:

- Protection of business data.
- Improved privacy.
- Better trust.

Confidentiality is one of the core principles of information security.

8.13 DATA INTEGRITY

Data integrity ensures that information remains accurate and consistent.

Methods Used:

- Input validation.
- Database constraints.
- Controlled updates.

Benefits:

- Reliable reports.
- Accurate analysis.
- Better decision-making.

Maintaining data integrity is essential for trustworthy analytics.

8.14 DATA AVAILABILITY

Availability ensures that data and services are accessible whenever required.

Methods Used:

- Stable database design.
- Efficient application architecture.
- Reliable deployment platform.

Benefits:

- Continuous access.
- Improved productivity.
- Better user experience.

Availability completes the CIA (Confidentiality, Integrity, Availability) security model.

8.15 SECURITY THREATS AND COUNTERMEASURES

Threat 1:

Unauthorized Access

Countermeasure:

Authentication System

Threat 2:

Duplicate User Creation

Countermeasure:

Username Validation

Threat 3:

Weak Passwords

Countermeasure:

Minimum Password Length Validation

Threat 4:

Data Manipulation

Countermeasure:

Role-Based Access Control

Threat 5:

Untracked Activities

Countermeasure:

Activity Logging

These countermeasures help improve overall system security.

8.16 ADVANTAGES OF IMPLEMENTED SECURITY MECHANISMS

- Improved data protection.
- Better access control.
- Enhanced accountability.
- Secure user management.
- Reduced security risks.
- Increased user trust.
- Better compliance with security standards.

8.17 LIMITATIONS OF CURRENT SECURITY SYSTEM

Although the project implements several security features, some limitations exist:

- Passwords are stored in a basic format.
- No two-factor authentication.
- No encryption mechanism.
- Limited advanced security controls.

These limitations provide opportunities for future improvement.

8.18 FUTURE SECURITY ENHANCEMENTS

The following security features can be implemented in future versions:

- 1.Password Hashing
Store passwords securely using hashing algorithms.
- 2.Two-Factor Authentication (2FA)
Add an additional layer of security.
- 3.Email Verification
Verify user identity through email.
- 4.Encryption
Encrypt sensitive information before storage.
- 5.Advanced Activity Monitoring
Track additional user activities.
- 6.Cloud Security Integration
Improve security for cloud-based deployments.
- 7.Automated Threat Detection
Identify suspicious activities automatically.

These enhancements will significantly strengthen system security.

8.19 SECURITY EVALUATION

The implemented security mechanisms successfully protect the application from common threats and unauthorized access.

Key Security Features Achieved:

- Authentication
- Authorization
- Role-Based Access Control
- Input Validation
- Session Management
- Activity Logging

The system demonstrates a satisfactory level of security for a student-level analytics application.

8.20 CHAPTER SUMMARY

This chapter discussed the security mechanisms implemented in the E-Commerce Sales Data Analytics Dashboard. Security features such as authentication, authorization, role-based access control, password validation, session management, activity logging, and database security were examined in detail. These mechanisms help protect business information, improve user accountability, and maintain data integrity. The chapter also discussed future security enhancements that can further strengthen the system and provide better protection against emerging threats.

CHAPTER 9

SCOPE AND FUTURE ENHANCEMENTS

9.1 INTRODUCTION

Every software system is developed to solve a specific problem and achieve predefined objectives. However, with changing business requirements and technological advancements, there is always scope for improvement and expansion. The E-Commerce Sales Data Analytics Dashboard has been designed to provide efficient sales analysis, business insights, and report generation. Although the current system successfully fulfills its intended objectives, several opportunities exist for future enhancements. This chapter discusses the current scope of the project, its practical applications, limitations, and future enhancements that can make the system more powerful, intelligent, and business-oriented.

9.2 CURRENT SCOPE OF THE PROJECT

The current scope of the E-Commerce Sales Data Analytics Dashboard includes the collection, processing, analysis, and visualization of sales data.

The developed system provides:

- User Authentication
- Role-Based Access Control
- Interactive Dashboard
- KPI Cards
- Sales Analysis
- Profit Analysis

- Region-wise Analysis
- Category-wise Analysis
- User Management
- Activity Logging
- PDF Report Generation
- Database Integration

The system enables organizations to monitor business performance efficiently and supports informed decision-making.

9.3 BUSINESS APPLICATIONS OF THE SYSTEM

The developed dashboard can be used in various business sectors.

9.3.1 Retail Industry

Retail businesses can use the dashboard to analyze product sales and customer preferences.

Benefits:

- Better inventory management
- Improved sales tracking
- Identification of best-selling products
- Enhanced business planning

9.3.2 E-Commerce Platforms

Online businesses can use the dashboard to evaluate sales performance.

Benefits:

- Real-time analytics
- Revenue monitoring
- Customer behavior analysis
- Marketing effectiveness evaluation

9.3.3 Small and Medium Enterprises (SMEs)

SMEs often lack advanced analytical systems due to budget constraints.

Benefits:

- Low-cost analytics solution
- Easy implementation

- Improved business monitoring
- Better profitability analysis

9.3.4 Educational Purposes

The dashboard can be used by students and researchers for learning data analytics concepts.

Benefits:

- Practical exposure
- Data visualization learning
- Business intelligence understanding
- Analytics project implementation

9.4 BENEFITS OF THE CURRENT SYSTEM

The developed system offers several benefits.

Operational Benefits:

- Faster analysis
- Reduced manual effort
- Improved productivity

Technical Benefits:

- Easy deployment
- Open-source technologies
- Lightweight architecture

Management Benefits:

- Better decision-making
- Accurate reporting
- Business performance monitoring

Financial Benefits:

- Cost-effective solution
- Improved profitability tracking
- Better resource utilization

9.5 LIMITATIONS OF THE CURRENT SYSTEM

Although the project successfully achieves its objectives, some limitations exist.

1.Static Dataset

The current system uses a predefined dataset.

Limitation:

Real-time sales updates are not available.

2.Limited User Roles

Only Admin and User roles are implemented.

Limitation:

Advanced permission levels are not available.

3.Basic Security Features

The project uses standard authentication mechanisms.

Limitation:

Advanced security controls are not implemented.

4.Local Database

SQLite is used as the database.

Limitation:

Not suitable for very large organizations.

5.Manual Dataset Upload

Data updates require manual intervention.

Limitation:

No automated synchronization.

9.6 FUTURE ENHANCEMENT OPPORTUNITIES

Technology evolves continuously, creating opportunities for future improvements.

The following enhancements can significantly improve the system.

9.7 REAL-TIME DATA ANALYTICS

Current Situation:

The dashboard analyzes static datasets.

Future Enhancement:

Implement real-time analytics using APIs and cloud databases.

Advantages:

- Live business monitoring
- Instant updates
- Better decision-making

Business Impact:

Managers can monitor sales performance in real time.

9.8 CLOUD DATABASE INTEGRATION

Current Situation:

SQLite database is used.

Future Enhancement:

Integrate cloud databases such as:

- MySQL
- PostgreSQL
- MongoDB
- Firebase

Benefits:

- Better scalability
- Improved accessibility
- Enhanced reliability
- Multi-user support

9.9 MACHINE LEARNING INTEGRATION

Machine Learning can significantly improve analytical capabilities.

Possible Applications:

1.Sales Prediction

Predict future sales based on historical data.

2.Profit Forecasting

Estimate future profitability.

3.Demand Forecasting

Predict product demand.

4.Customer Segmentation

Group customers according to purchasing behavior.

Benefits:

- Intelligent insights
- Better planning
- Improved forecasting

9.10 ARTIFICIAL INTELLIGENCE IMPLEMENTATION

Artificial Intelligence can enhance the dashboard by providing smart recommendations.

AI Features:

- Automated Insights
- Recommendation Engine
- Intelligent Reporting
- Customer Behavior Analysis

Benefits:

- Reduced manual effort
- Smarter decisions
- Better business strategies

9.11 CUSTOMER ANALYTICS MODULE

Future versions can include advanced customer analytics.

Features:

- Customer Lifetime Value
- Purchase Frequency Analysis
- Customer Retention Analysis
- Customer Segmentation

Benefits:

- Improved customer relationships
- Better marketing strategies
- Increased customer satisfaction

9.12 INVENTORY MANAGEMENT INTEGRATION

Inventory management can be integrated with sales analytics.

Features:

- Stock Monitoring
- Inventory Alerts
- Product Availability Tracking

Benefits:

- Reduced stock shortages
- Improved inventory planning
- Better supply chain management

9.13 MOBILE APPLICATION DEVELOPMENT

Current Situation:

System is web-based.

Future Enhancement:

Develop Android and iOS applications.

Benefits:

- Mobile accessibility
- Better user convenience
- Increased flexibility

Business Impact:

Managers can monitor business performance from anywhere.

9.14 EMAIL REPORT AUTOMATION

Future versions can automatically send reports through email.

Features:

- Daily Reports
- Weekly Reports
- Monthly Reports

Benefits:

- Time saving
- Automated communication
- Better information sharing

9.15 ADVANCED DATA VISUALIZATION

Future versions can include:

- Interactive Dashboards
- Advanced Charts
- Heat Maps
- Geographic Maps
- Drill-Down Analytics

Benefits:

- Better understanding
- Improved presentation
- Enhanced user experience

9.16 MULTI-USER SUPPORT

Current Situation:

Limited multi-user functionality.

Future Enhancement:

Develop enterprise-level multi-user access.

Benefits:

- Team collaboration
- Shared dashboards
- Improved management

9.17 ADVANCED SECURITY FEATURES

Future security improvements may include:

- Password Hashing
- Data Encryption
- Two-Factor Authentication
- Biometric Authentication
- Security Monitoring

Benefits:

- Better protection
- Improved privacy
- Reduced security risks

9.18 API INTEGRATION

Application Programming Interfaces (APIs) can be integrated with:

- Amazon
- Flipkart
- Shopify
- WooCommerce
- Payment Gateways

Benefits:

- Automated data collection
- Real-time synchronization
- Better analytics

9.19 BIG DATA ANALYTICS

Future versions can handle large-scale datasets using:

- Hadoop
- Spark
- Data Warehousing

Benefits:

- Large-scale analysis
- Better performance
- Enterprise-level capabilities

9.20 OVERALL FUTURE VISION

The long-term vision of the E-Commerce Sales Data Analytics Dashboard is to transform it into a complete Business Intelligence platform capable of:

- Real-Time Analytics
- AI-Based Decision Support
- Cloud Computing Integration
- Enterprise Resource Planning Support
- Predictive Analytics
- Advanced Visualization
- Automated Reporting

Such enhancements will make the system suitable for organizations of all sizes.

9.21 CHAPTER SUMMARY

This chapter discussed the scope and future enhancement opportunities of the E-Commerce Sales Data Analytics Dashboard. The current scope includes sales analysis, report generation, user management, and data visualization. Several future improvements such as Artificial Intelligence, Machine Learning, Real-Time Analytics, Cloud Integration, Mobile Applications, Advanced Security, and API Integration were identified. These enhancements can significantly increase the functionality, scalability, and business value of the system in future versions.

CHAPTER 10

CONCLUSION

10.1 INTRODUCTION

The successful completion of a software project is determined by its ability to meet user requirements and solve the identified problem effectively. The E-Commerce Sales Data Analytics Dashboard was developed with the objective of transforming raw sales data into meaningful business insights through data analysis, visualization, and reporting. The project combines the concepts of data analytics, business intelligence, database management, and web application development to provide a practical solution for monitoring and evaluating sales performance.

This project demonstrates how modern analytical tools can help organizations make better decisions by presenting complex data in a simple and understandable format. The developed system provides an interactive environment where users can easily access important business information and generate reports without requiring advanced technical knowledge.

10.2 PROJECT SUMMARY

The E-Commerce Sales Data Analytics Dashboard was designed and implemented to address the challenges associated with manual sales analysis. Traditional methods often require significant time and effort to process large volumes of sales records, making it difficult for organizations to obtain timely insights. The proposed system eliminates many of these challenges by automating data processing and presenting information through interactive dashboards.

The system was developed using Python as the programming language, Streamlit as the web framework, SQLite as the database, Pandas for data processing, Plotly for data visualization, and ReportLab for report generation. These technologies were selected because they are reliable, efficient, and widely used in modern data analytics applications.

Throughout the development process, various modules were implemented, including user authentication, role-based access control, dashboard visualization, user management, activity logging, and PDF report generation. Together, these modules form a complete analytics platform capable of supporting business decision-making.

10.3 ACHIEVEMENT OF PROJECT OBJECTIVES

The primary objective of the project was to develop an analytics dashboard capable of processing sales data and presenting valuable insights in an easy-to-understand format. The developed system successfully achieved this objective by providing interactive visualizations, analytical reports, and performance indicators.

The project also aimed to improve security by implementing authentication and role-based access control mechanisms. This objective was achieved through a login system that restricts access to authorized users and provides different permissions based on user roles.

Another important objective was to automate report generation. The system successfully generates PDF reports containing sales summaries and business performance information. This feature reduces manual effort and allows users to share analytical findings efficiently.

The project further achieved its goal of improving business analysis by providing category-wise analysis, region-wise analysis, profit analysis, and sales trend analysis. These features help users understand business performance and identify areas for improvement.

10.4 SIGNIFICANCE OF THE PROJECT

The significance of this project lies in its ability to convert raw business data into actionable information. In today's competitive business environment, organizations must make decisions based on accurate and timely information. The dashboard developed in this project provides a practical solution for achieving this objective.

The project demonstrates the importance of data analytics in modern organizations. Instead of manually analyzing spreadsheets, users can access visual reports and dashboards that simplify decision-making. This not only improves efficiency but also reduces the possibility of human error.

Furthermore, the project provides valuable learning opportunities in software development, database management, web application design, and business intelligence. The knowledge gained during the development process can be applied to future projects involving analytics and decision-support systems.

10.5 PRACTICAL BENEFITS OF THE SYSTEM

The developed dashboard offers several practical benefits to users and organizations. By providing a centralized platform for sales analysis, the system enables managers to monitor business performance more effectively. Users can quickly identify profitable products, high-performing regions, and important sales trends without performing complex calculations.

The dashboard also improves productivity by reducing the time required to prepare reports. Information that previously required manual analysis can now be generated automatically through charts and downloadable reports. This allows organizations to focus more on strategic planning and less on data preparation. Another important benefit is improved data accessibility. The dashboard presents information through interactive visualizations that are easy to understand, even for users without technical expertise. This increases the usability and effectiveness of the analytical process.

10.6 CHALLENGES FACED DURING DEVELOPMENT

During the development of the project, several challenges were encountered. One of the primary challenges involved database integration and management. Ensuring that user information, authentication details, and activity logs were stored and retrieved correctly required careful implementation and testing. Another challenge involved session management and user authentication. Maintaining secure login sessions while ensuring smooth navigation between different sections of the dashboard required additional development effort. The implementation of PDF report generation also presented difficulties. Formatting reports correctly and ensuring that analytical information was displayed properly required multiple revisions and testing cycles. Despite these challenges, systematic debugging, testing, and validation enabled the successful completion of the project. The experience gained while solving these problems contributed significantly to the overall learning process.

10.7 LEARNING OUTCOMES

The development of this project provided valuable technical and practical knowledge. It offered hands-on experience in working with Python programming, database management, data processing libraries, and web application frameworks.

The project also enhanced understanding of data analytics concepts such as data visualization, business intelligence, reporting systems, and performance analysis. Practical exposure to tools such as Streamlit, Pandas, Plotly, SQLite, and ReportLab improved technical skills and provided insight into real-world software development practices.

In addition to technical knowledge, the project strengthened problem-solving abilities, project planning skills, testing methodologies, and documentation techniques. These skills are valuable for future academic and professional activities.

10.8 FUTURE POTENTIAL

Although the current system successfully fulfills its objectives, significant opportunities exist for future enhancement. The integration of artificial intelligence and machine learning can enable predictive analytics and sales forecasting capabilities. Cloud database integration can improve scalability and accessibility, while mobile application development can increase convenience for users.

Future versions may also incorporate advanced security features such as password hashing, encryption, and two-factor authentication. Real-time data synchronization and API integration can further enhance the functionality of the system and expand its practical applications.

These enhancements would transform the dashboard from a student-level analytics project into a more comprehensive business intelligence platform suitable for larger organizations.

10.9 FINAL CONCLUSION

The E-Commerce Sales Data Analytics Dashboard successfully demonstrates the application of data analytics and business intelligence concepts in solving real-world business problems. The system provides an effective platform for analyzing sales performance, generating reports, and supporting decision-making processes.

The project achieved all major objectives identified during the planning and analysis stages. It successfully integrates user authentication, data processing, visualization, report generation, and database management into a single application. The resulting system is reliable, user-friendly, and capable of

providing meaningful business insights.

Overall, the project represents a successful implementation of modern analytics techniques and serves as a valuable example of how technology can be used to improve organizational efficiency and decision-making. The knowledge gained through the development process has contributed significantly to both technical and professional growth, making this project an important academic achievement and a strong foundation for future work in the fields of data analytics, business intelligence, and software development.

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- 2.Streamlit Official Documentation.
- 3.Pandas User Guide.

4. Plotly Visualization Documentation.
5. SQLite Documentation.
6. Data Analytics and Business Intelligence Research Papers.

Project Resources

1. E-Commerce Sales Dataset used for analysis.
2. GitHub Repository of the Project.
3. Streamlit Deployment Platform.
4. Project Design Documents.
5. Project Testing Reports.

Appendix A – Source Code

- app.py

```
import streamlit as st
import pandas as pd
import plotly.express as px
import sqlite3
from io import BytesIO

from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table,
TableStyle
from reportlab.lib import colors
from datetime import datetime
from reportlab.lib.styles import getSampleStyleSheet

import init_db
import insert_users

from database import authenticate
from logger import save_log
from backend import *

# Initialize session state
if "logged_in" not in st.session_state:
    st.session_state.logged_in = False

if "role" not in st.session_state:
    st.session_state.role = None

def generate_pdf(total_sales, total_profit, total_orders):

    pdf = SimpleDocTemplate("sales_report.pdf")

    styles = getSampleStyleSheet()

    elements = []
```

Title

```
elements.append(  
    Paragraph(  
        "E-Commerce Sales Analytics Dashboard Report",  
        styles["Title"]  
    )  
)
```

```
elements.append(Spacer(1, 20))
```

Date

```
elements.append(  
    Paragraph(  
        f"Generated On: {datetime.now().strftime('%d-%m-%Y %H:%M')}",  
        styles["Normal"]  
    )  
)
```

```
elements.append(Spacer(1, 12))
```

Executive Summary

```
elements.append(  
    Paragraph(  
        "Executive Summary",  
        styles["Heading2"]  
    )  
)
```

```
elements.append(  
    Paragraph(  
        "This report provides an overview of sales performance, profit analysis, "  
        "and order statistics generated from the E-Commerce Analytics  
Dashboard.",  
        styles["Normal"]  
    )  
)
```

```
elements.append(Spacer(1, 15))
```

Dashboard Highlights

```
elements.append(  
    Paragraph(  
        "Dashboard Highlights",  
        styles["Heading2"]  
    )  
)  
  
elements.append(  
    Paragraph(  
        "• Sales Performance Tracking<br/>"  
        "• Profit Analysis<br/>"  
        "• Order Monitoring<br/>"  
        "• Customer Insights<br/>"  
        "• Region-wise Analysis",  
        styles["Normal"]  
    )  
)  
  
elements.append(Spacer(1, 15))
```

KPI Table

```
data = [  
    ["Metric", "Value"],  
    ["Total Sales", f"₹ {total_sales:,.2f}"],  
    ["Total Profit", f"₹ {total_profit:,.2f}"],  
    ["Total Orders", str(total_orders)]  
]  
  
table = Table(data, colWidths=[220, 180])  
  
table.setStyle(TableStyle([  
    ('BACKGROUND', (0, 0), (-1, 0), colors.grey),  
    ('TEXTCOLOR', (0, 0), (-1, 0), colors.whitesmoke),  
    ('GRID', (0, 0), (-1, -1), 1, colors.black),  
    ('BACKGROUND', (0, 1), (-1, -1), colors.beige),  
    ('ALIGN', (0, 0), (-1, -1), 'CENTER')  
]))
```

```
elements.append(table)

elements.append(Spacer(1, 20))

# Conclusion
elements.append(
    Paragraph(
        "Conclusion",
        styles["Heading2"]
    )
)

elements.append(
    Paragraph(
        "The dashboard helps businesses analyze sales trends, monitor profits, "
        "and track order activity. These insights support better decision-making "
        "and business growth.",
        styles["Normal"]
    )
)

elements.append(Spacer(1, 20))

# Footer
elements.append(
    Paragraph(
        "Generated using E-Commerce Sales Analytics
Dashboard",
        styles["Italic"]
    )
)

pdf.build(elements)

return "sales_report.pdf"

# ----- LOGIN SYSTEM -----
```

```
def login():

    st.title("🔒 E-Commerce Dashboard Login")

    username = st.text_input("Username")
    password = st.text_input("Password", type="password")

    if st.button("Login"):

        success, role = authenticate(username, password)

        if success:
            st.session_state.logged_in = True
            st.session_state.role = role
            st.session_state.username = username

            save_log(username, "Login")

            st.success("Login Successful")
            st.rerun()

        else:
            st.error("Invalid Username or Password")

# Session state
if "username" not in st.session_state:
    st.session_state.username = ""

if "role" not in st.session_state:
    st.session_state.role = ""

# Initialize session state
if "logged_in" not in st.session_state:
    st.session_state.logged_in = False

if "role" not in st.session_state:
    st.session_state.role = None

# Show Login page if not Logged in
```

```
if not st.session_state.logged_in:
    login()
    st.stop()

# Page Config
st.set_page_config(page_title="E-Commerce Analytics",
                    layout="wide")

# Custom CSS (Premium UI)
st.markdown("""
    <style>
    body {
        background-color: #0E1117;
        color: white;
    }
    .stMetric {
        background-color: #1c1f26;
        padding: 15px;
        border-radius: 10px;
    }
    </style>
    """, unsafe_allow_html=True)

# Title
st.title("🛒 E-Commerce Sales Analytics Dashboard")

# Logout Button
if st.button("🚪 Logout"):

    save_log(st.session_state.username, "Logout")

    st.session_state.logged_in = False
    st.rerun()

# Load Data
df = load_data()

# Sidebar Filters
st.sidebar.header("🔍 Filters")
```

```
region_filter = st.sidebar.multiselect(
    "Select Region",
    options=df["Region"].unique(),
    default=df["Region"].unique()
)

category_filter = st.sidebar.multiselect(
    "Select Category",
    options=df["Category"].unique(),
    default=df["Category"].unique()
)

# Filter Data
filtered_df = df[
    (df["Region"].isin(region_filter)) &
    (df["Category"].isin(category_filter))
]

# KPIs
total_sales_value = total_sales(filtered_df)
total_profit_value = total_profit(filtered_df)
total_orders_value = total_orders(filtered_df)

col1, col2, col3 = st.columns(3)

col1.metric("💰 Total Sales", total_sales_value)
col2.metric("📈 Total Profit", total_profit_value)
col3.metric("📦 Total Orders", total_orders_value)

# Sales Trend
st.subheader("📅 Monthly Sales Trend")
filtered_df["OrderDate"] =
pd.to_datetime(filtered_df["OrderDate"])
monthly =
filtered_df.groupby(filtered_df["OrderDate"].dt.to_period(
    "M"))["Sales"].sum().reset_index()
monthly["OrderDate"] = monthly["OrderDate"].astype(str)
```

```
fig1 = px.line(monthly, x="OrderDate", y="Sales",
title="Monthly Sales")
st.plotly_chart(fig1, use_container_width=True)

# Category Sales
col4, col5 = st.columns(2)

with col4:
    st.subheader("📊 Category Sales")
    cat =
filtered_df.groupby("Category")["Sales"].sum().reset_index()
    fig2 = px.bar(cat, x="Category", y="Sales",
color="Category")
    st.plotly_chart(fig2, use_container_width=True)

with col5:
    st.subheader("🌐 Region Sales")
    reg =
filtered_df.groupby("Region")["Sales"].sum().reset_index()
    fig3 = px.pie(reg, names="Region", values="Sales")
    st.plotly_chart(fig3, use_container_width=True)

# Top Products
st.subheader("🔥 Top Products")
top_products =
filtered_df.groupby("Product")["Sales"].sum().sort_values(as
cending=False).head(10)
st.dataframe(top_products)

# Customer Analysis
st.subheader("👤 Top Customers")
customers =
filtered_df.groupby("Customer")["Sales"].sum().sort_values(a
scending=False).head(10)
st.dataframe(customers)

# Full Data
st.subheader("📁 Full Dataset")
st.dataframe(filtered_df)
```

```
st.subheader("📄 Download Reports")

csv = filtered_df.to_csv(index=False)

st.download_button(
    label="Download Sales Report",
    data=csv,
    file_name="sales_report.csv",
    mime="text/csv"
)

st.write("Sales:", total_sales_value)
st.write("Profit:", total_profit_value)
st.write("Orders:", total_orders_value)

pdf_file = generate_pdf(
    total_sales_value,
    total_profit_value,
    total_orders_value
)

with open(pdf_file, "rb") as f:
    st.download_button(
        label="Download PDF Report",
        data=f,
        file_name="sales_report.pdf",
        mime="application/pdf"
    )

if st.session_state.role == "admin":

    st.subheader("🔑 Admin Panel")
    conn = sqlite3.connect("database.db")
    cursor = conn.cursor()

    cursor.execute("SELECT COUNT(DISTINCT username) FROM Users")
    total_users = cursor.fetchone()[0]
```

```
conn.close()

st.metric("👤 Total Users", total_users)

st.subheader("✚ Add New User")

new_username = st.text_input("New Username")
new_password = st.text_input("New Password",
type="password")
new_role = st.selectbox("Role", ["user", "admin"])

if st.button("Create User"):

    if new_username.strip() == "":
        st.error("Username cannot be empty")

    elif len(new_password) < 6:
        st.error("Password must be at least 6
characters")

    else:

        conn = sqlite3.connect("database.db")
        cursor = conn.cursor()

        cursor.execute(
            "SELECT * FROM Users WHERE username=?",
            (new_username,)
        )

        existing_user = cursor.fetchone()

        if existing_user:
            st.error("Username already exists")

        else:
            cursor.execute(
```

```
        "INSERT INTO Users (username, password,
role) VALUES (?, ?, ?)",
        (new_username, new_password, new_role)
    )

    conn.commit()
    st.success("User Created Successfully ✔")

    conn.close()

st.subheader("👤 Users Database")

conn = sqlite3.connect("database.db")

users_df = pd.read_sql_query(
    "SELECT username, role FROM Users",
    conn
)

st.dataframe(users_df)

conn.close()

logs = pd.read_csv("logs.csv")

st.write("User Activity Logs")

logs = pd.read_csv("logs.csv")    # Read latest Logs
every time
st.dataframe(logs)
```

- database.py

```
import sqlite3
```

```
def authenticate(username, password):

    conn = sqlite3.connect("database.db")
    cursor = conn.cursor()

    cursor.execute(
        "SELECT role FROM Users WHERE username=? AND
password=?",
        (username, password)
    )

    user = cursor.fetchone()

    conn.close()

    if user:
        return True, user[0]    # success, role
    else:
        return False, None
```

```
def create_user(username, password, role):

    conn = sqlite3.connect("database.db")
    cursor = conn.cursor()

    cursor.execute(
        "SELECT * FROM Users WHERE username=?",
        (username,)
    )

    existing_user = cursor.fetchone()

    if existing_user:
        conn.close()
        return False
```

```
cursor.execute(
    "INSERT INTO Users(username,password,role) VALUES\n",
    (username, password, role)
)

conn.commit()
conn.close()

return True
```

- logger.py

```
import pandas as pd
from datetime import datetime

def save_log(username, action):

    try:
        logs = pd.read_csv("logs.csv")
    except:
        logs =
pd.DataFrame(columns=["username","action","time"])

    new_log = pd.DataFrame({
        "username":[username],
        "action":[action],
        "time":[datetime.now()]
    })

    logs = pd.concat([logs,new_log])

    logs.to_csv("logs.csv",index=False
```

- insert_users.py

```
import sqlite3

conn = sqlite3.connect("database.db")
cursor = conn.cursor()

users = [
    ("admin", "1234", "admin"),
    ("noor", "1234", "user")
]

for user in users:
    cursor.execute(
        "SELECT * FROM Users WHERE username=?",
        (user[0],)
    )

    if cursor.fetchone() is None:
        cursor.execute(
            "INSERT INTO Users (username,password,role)
VALUES (?, ?, ?)",
            user
        )

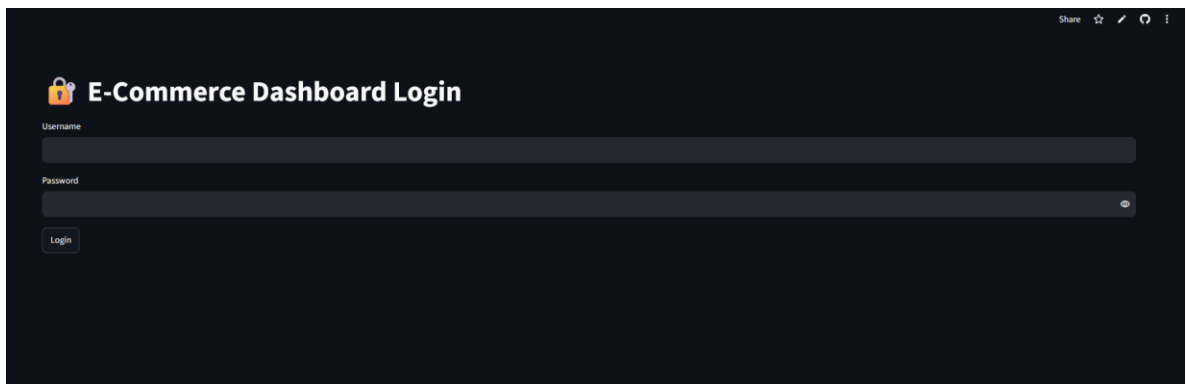
conn.commit()
conn.close()

print("Users Added")
```

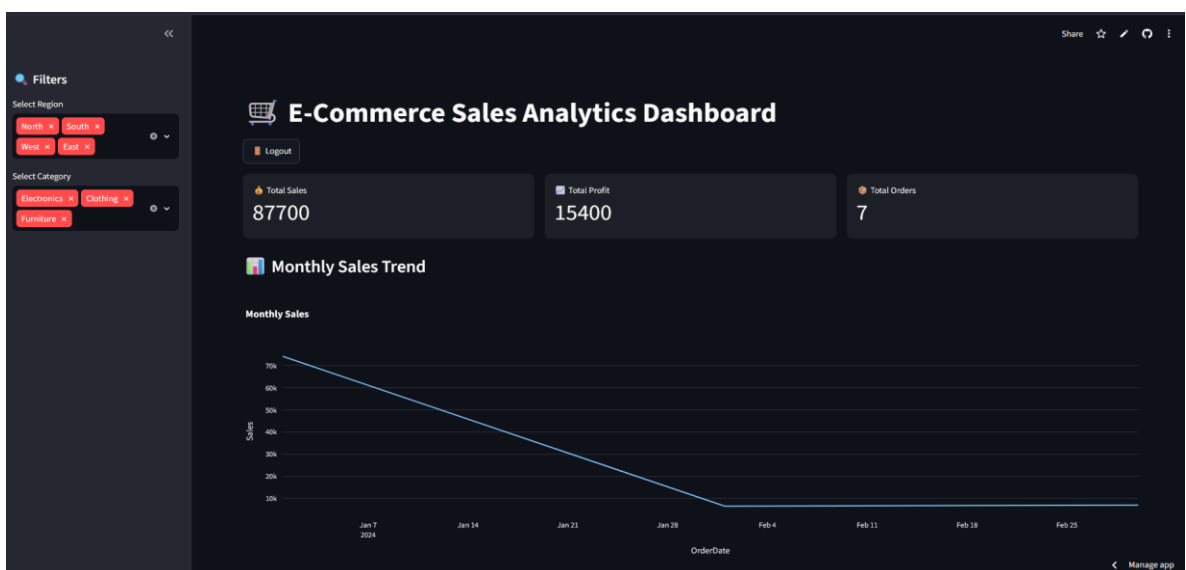
Appendix B – Screenshots

1. Login Page
2. Dashboard
3. KPI Cards
4. Sales Trend Analysis
5. Category Analysis
6. Region Analysis
7. Admin Panel
8. Activity Logs
9. PDF Report

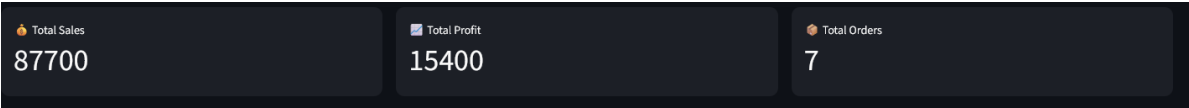
1.



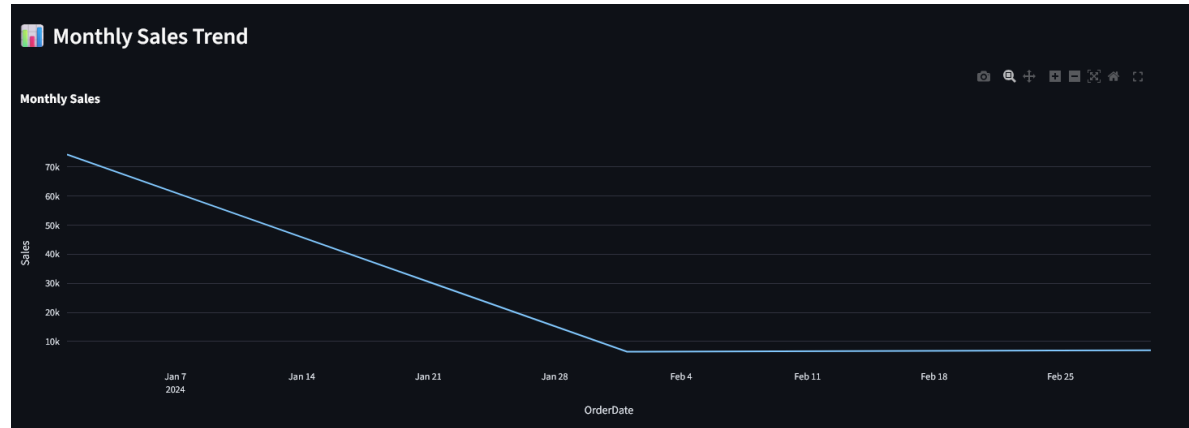
2.



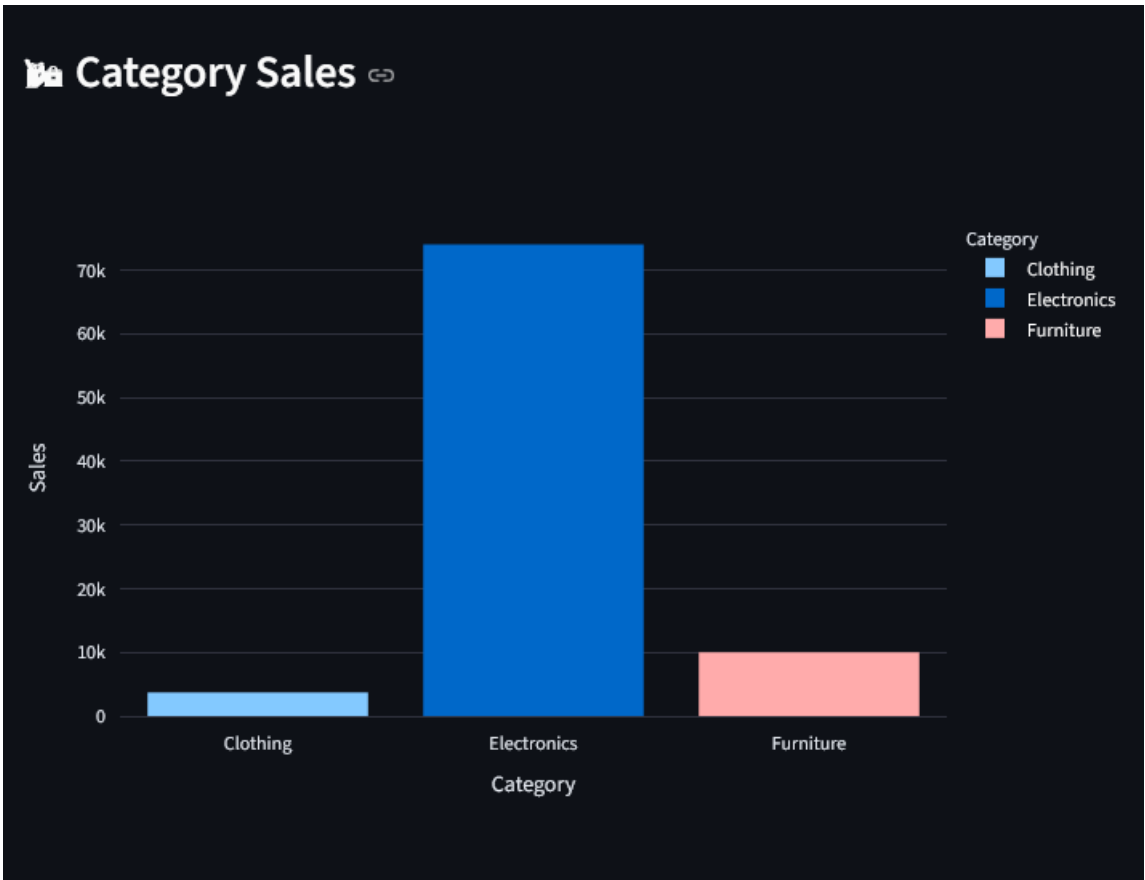
3.



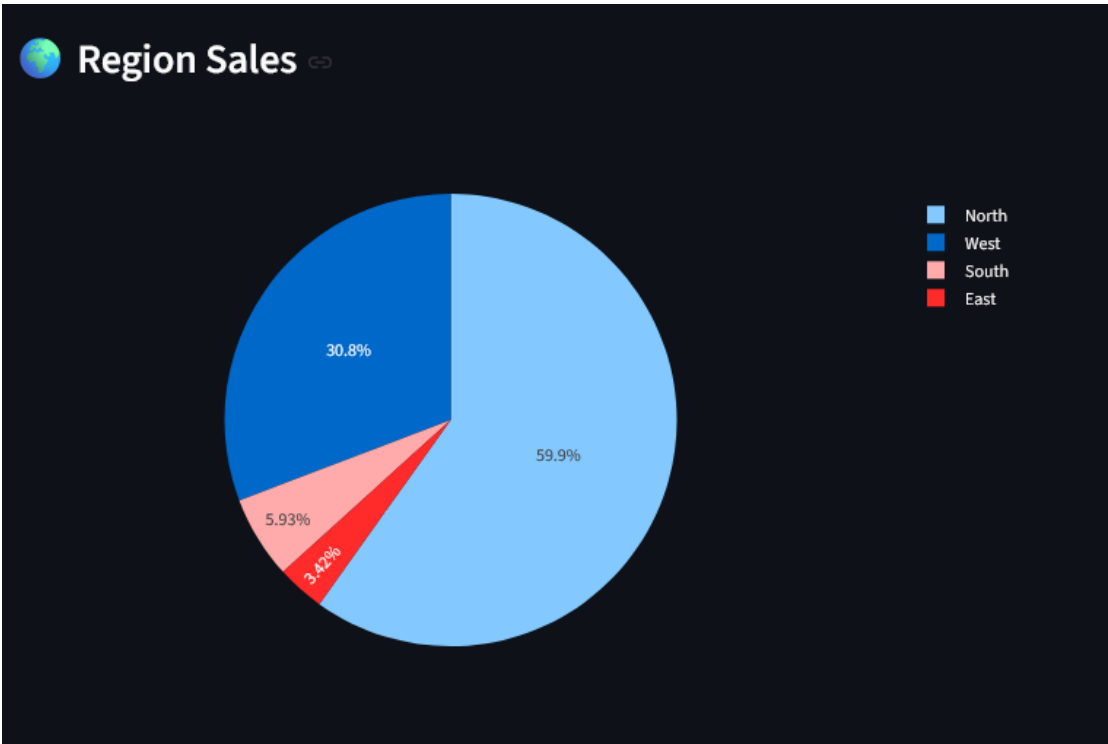
4.



5.



6.



7.

 **Admin Panel**

 Total Users

2

 **Add New User**

New Username

New Password

Role

user

Create User

Password must be at least 6 characters

8.

User Activity Logs

	username	action	time
0	admin	Login	2026-07-17 11:06:29.763233
1	admin	Logout	2026-07-17 13:02:59.021501
2	admin	Login	2026-07-17 13:05:29.046922

9.

E-Commerce Sales Analytics Dashboard Report

Generated On: 17-07-2026 13:05

Executive Summary

This report provides an overview of sales performance, profit analysis, and order statistics generated from the E-Commerce Analytics Dashboard.

Dashboard Highlights

- Sales Performance Tracking
- Profit Analysis
- Order Monitoring
- Customer Insights
- Region-wise Analysis

Metric	Value
Total Sales	■ 87,700.00
Total Profit	■ 15,400.00
Total Orders	7

Conclusion

The dashboard helps businesses analyze sales trends, monitor profits, and track order activity. These insights support better decision-making and business growth.

Generated using E-Commerce Sales Analytics Dashboard

APPENDIX C

USER MANUAL

C.1 Introduction

The E-Commerce Sales Data Analytics Dashboard is designed to help users analyze sales performance through interactive dashboards and reports. This manual explains how to use the system effectively.

C.2 System Requirements

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: 4 GB or higher
- Storage: 500 MB free space

Software Requirements

- Windows 10/11
- Python 3.x
- Streamlit
- SQLite

C.3 Starting the Application

Step 1:

Open Command Prompt or Terminal.

Step 2:

Navigate to the project folder.

Step 3:

Execute the command:

```
streamlit run app.py
```

Step 4:

The application will open in a web browser.

C.4 Login Procedure

1. Enter Username.
2. Enter Password.
3. Click Login.
4. Dashboard will open after successful authentication.

C.5 Dashboard Usage

The dashboard displays:

- Total Sales
- Total Profit
- Total Orders
- Category Analysis
- Region Analysis
- Sales Trends

Users can view charts and reports for business analysis.

C.6 User Management

Administrator Functions:

- Create New Users
- View Existing Users
- Manage User Roles

Validation Rules:

- Username cannot be empty.
- Password must contain at least six characters.
- Duplicate usernames are not allowed.

C.7 Activity Logs

Administrators can view login and logout activities of users.

C.8 Report Generation

Steps:

- 1.Open Dashboard.
- 2.Click Generate Report.
- 3.Download PDF Report.

C.9 Logout Procedure

- 1.Click Logout Button.
- 2.Session closes successfully.
- 3.User is redirected to Login Page.

C.10 Troubleshooting

Problem:

Login Failed

Solution:

Verify username and password.

Problem:

Database Error

Solution:

Check database connection.

Problem:

Report Not Downloading

Solution:

Verify report generation module.

C.11 Conclusion

The system is designed to be simple, secure, and user-friendly. Users can analyze business data efficiently and generate reports for decision-making purposes